

Adams-type maps and transfinite factorizations of ring spectra

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Abstract

We study three questions about Adams-type maps of ring spectra. We construct a descent-flat associative map which is not Adams-type, although its comodule-valued homology theory is adapted. We give an associative map which admits no continuous transfinite factorization into descent-flat maps, and show that the zero augmentation of the free commutative sphere algebra on a degree-zero generator admits no such factorization into Adams-type commutative maps. The latter obstruction uses primitive classes in bar homology, a Hopf structure on the relative bar filtration, and descent along coideal subalgebras. We also give an Adams-type map of ordinary commutative rings and a module with free base-change homology which cannot be expressed as a filtered colimit of perfect modules with projective base-change homology. Positive results hold for almost perfect modules over connective sources with discrete coherent targets, and for maps represented by strict commutative differential graded algebras.

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1 Introduction

For a map of ring spectra $A \rightarrow B$, the Adams-type condition asks for a perfect approximation to B whose terms become projective after base change to B . It is stronger than flatness of the cooperation module $\pi_*(B \otimes_A B)$. Burklund, Levy, and Pstrągowski showed that neither condition is stable under composition [BLP, Theorem 2]. They also proved that a map of commutative ring spectra whose target admits an algebra structure over a field is a transfinite composite of Adams-type maps [BLP, Theorem 6]. Their three concluding questions concern the difference between the two flatness conditions, the existence of such factorizations in general, and the corresponding approximation problem for modules. We address these questions in their unrestricted, possibly nonconnective form.

Definition 1.1. A map $A \rightarrow B$ of associative ring spectra is *Adams-type* if, as a left A -module, B is a filtered colimit of perfect modules P_α for which $\pi_*(B \otimes_A P_\alpha)$ is projective over the graded ring π_*B . It is *descent-flat* if $\pi_*(B \otimes_A B)$ is flat as a left π_*B -module. A perfect A -module with the displayed projectivity property will be called *B -good*.

All tensor products of spectra are derived. A continuous transfinite composite is indexed by a small ordinal, with the value at each nonzero limit ordinal equal to the colimit of the earlier values. The category in which the intermediate maps lie is part of the assertion.

The questions can now be stated in terms of the same approximation condition:

- (1) Does descent-flatness imply the Adams-type condition?
- (2) Does every ring map admit a continuous transfinite factorization into Adams-type maps, in the associative or in the commutative category?
- (3) If $A \rightarrow B$ is Adams-type and $\pi_*(B \otimes_A M)$ is flat, must M be a filtered colimit of B -good perfect modules?

We retain the numbering of [BLP]. Theorems A, B, and C address these three questions, respectively.

1.1 Counterexamples

The first result separates descent-flatness from Adams-type approximation, including the adapted variant in the footnote to [BLP, Question (1)]. The definition of adapted used here is recalled in Section 3.

Theorem A. *There is a map $A \rightarrow HR$, where $R = \mathbb{F}_2[\varepsilon]/(\varepsilon^2)$ and A is an associative ring spectrum, such that:*

- (i) $\pi_*(HR \otimes_A HR)$ is free on both sides over R ;
- (ii) the associated comodule-valued homology theory on Mod_A is adapted;
- (iii) the map is not Adams-type.

The source has an explicit differential graded model. A natural square-zero operator on base-change homology sends an infinite sequence of basis vectors to their successors multiplied by ε . In a projective R -module, the identity $\text{ann}(\varepsilon) = \varepsilon(-)$ forces any lift of the first basis vector, together with one finite relation, to contain the whole sequence. This contradicts finite generation for a B -good perfect module. The proof also computes the canonical coalgebra comparison required for adaptedness. Section 3 gives a finite algebra presentation variant and a variant with connective source and periodic target.

For the factorization question the associative and commutative counterexamples use different arguments.

Theorem B. (i) *For every prime p , the diagonal inclusion*

$$H\mathbb{Z} \times H\mathbb{F}_p \longrightarrow H \begin{pmatrix} \mathbb{Z} & \mathbb{F}_p \\ 0 & \mathbb{F}_p \end{pmatrix}$$

is not a continuous transfinite composite of descent-flat maps of associative ring spectra.

- (ii) *The zero augmentation*

$$\text{Sym}_{\mathbb{S}}(\mathbb{S}) \longrightarrow \mathbb{S}$$

is not a continuous transfinite composite of Adams-type maps of commutative ring spectra.

In both assertions the intermediate ring spectra may be arbitrarily nonconnective.

For the first assertion, we preserve the vanishing of an entire bimodule mapping spectrum against every left-flat test object. The initial vanishing is forced by the two characteristics at the vertices. A module bar resolution propagates it through a descent-flat step; a limit description of module actions propagates it through a limit ordinal. A two-term calculation at the triangular target contradicts this vanishing.

For the second assertion, the obstruction is a primitive bar-homology class whose powers are all nonzero. The main input is Theorem 8.1: an augmented commutative Adams step preserves this property for a homogeneous primitive of nonzero degree. Starting with a degree-one class in the bar of $\text{Sym}_{\mathbb{S}}(\mathbb{S})$, we can therefore follow it through a proposed factorization, although no such class exists at its endpoint.

To prove the single-step statement, the sphere augmentation first makes the relative algebra sphere-flat, forcing its nonzero-index Dyer–Lashof operations to vanish. The Hopf structure on the relative bar filtration expresses each successive zero-column kernel as an ideal generated by a coideal subalgebra. Faithfully flat descent then carries the operation constraint through the finite pages and prevents these kernels from killing a nonnilpotent

primitive. Finally, a class in the initial filtration stage that vanishes in the target already vanishes in a finite skeleton. This detects a contradiction on a finite page, without a strong-convergence assumption.

The module approximation question already fails over ordinary commutative rings.

Theorem C. *Let $R = \mathbb{F}_2[\varepsilon]/(\varepsilon^2)$. The quotient*

$$H(R[y]) \longrightarrow HR, \quad y \longmapsto 0,$$

is Adams-type. There is an $H(R[y])$ -module M for which $\pi_(HR \otimes_{H(R[y])} M)$ is free over R , but M is not a filtered colimit of HR -good perfect modules.*

Here M is represented by a complex with one generator m_i in degree $-i$ and differential $dm_i = \varepsilon y m_{i+1}$. A four-cell perfect complex records the relation needed by the operator obstruction. Compactness of this one witness excludes every filtered presentation, including presentations with noninjective transition maps. The periodic variant makes both the source and the module connective; its target remains nonconnective.

1.2 Positive results and scope

The same examples also delimit several positive statements. The descent-flat non-Adams maps used for Theorem A have two-step associative Adams factorizations; their shortest finite factorization length is exactly two. They therefore do not provide the obstruction in Theorem B.

If A is connective, S is a discrete coherent commutative ring, and an almost perfect A -module M has flat $\pi_*(HS \otimes_A M)$, then M does have a filtered presentation by HS -good perfect modules. This assertion does not require $A \rightarrow HS$ to be Adams-type. Its proof uses finite free simplicial resolutions and split cycle and boundary sequences over S .

Maps represented by strict commutative differential graded algebras also admit countable Adams factorizations. Here strict commutativity includes the relation that each odd-degree element has square zero. Each stage adds variables whose differentials lie in the preceding stage; finite monomial submodules become free after the change of variables $x' \mapsto x' - x$. These strict models form a specified subclass of commutative ring spectra. This construction does not supply a strict model for an arbitrary commutative ring spectrum.

The approximation results can be read through Sections 3, 4, and 11. The associative factorization arguments occupy Sections 5 and 10. The commutative obstruction is developed in Sections 6–9; its principal intermediate statement is Theorem 8.1.

2 Conventions and projective base change

We work in the infinity-categories of spectra, associative ring spectra, and commutative ring spectra. We also use the notations $\mathbb{E}_1$ and $\mathbb{E}_\infty$ for the last two types of algebra structure. Modules are left modules unless a side is specified. The notation $\text{Bimod}(A, T)$ means left A , right T bimodules with commuting coherent actions. The symbol Map denotes a mapping spectrum; brackets denote its degree-zero homotopy group.

All homotopy groups and modules over homotopy rings are graded. A module is *homotopy-flat* or *homotopy-projective* when its homotopy groups have that property over the relevant graded homotopy ring. No connectivity is implicit. An A -module is perfect precisely when it is compact, equivalently when it belongs to the closure of A under finite colimits, shifts, and retracts [HA, Proposition 7.2.4.2]. Every A -module is a filtered colimit of perfect modules.

Lemma 2.1. *An A -module P is homotopy-projective if and only if it is a retract of a coproduct of shifts of A . If P is also perfect, it is a retract of a finite such coproduct. In particular, π_*P is then finitely generated projective.*

Proof. Choose a graded free module surjecting onto π_*P , realize its generators by a map $W = \bigoplus_j \Sigma^{d_j} A \rightarrow P$, and split the corresponding graded idempotent on π_*W . Maps out of the shifts of A realize its matrix entries, so this idempotent is a homotopy idempotent of W . Splitting it in Mod_A gives a retract P' of W mapping to P by an isomorphism on homotopy groups, hence by an equivalence. The converse follows by taking homotopy groups. If P is compact, its inclusion in W factors through a finite subcoproduct; the resulting composite with $W \rightarrow P$ is still the identity. This gives the final assertion. $\square$

Thus the projectivity condition in Definition 1.1 can be checked on actual module retracts, as in [BLP, Lemma 3, opening observation]. Base change preserves these retract presentations and preserves perfect modules. Adams-type maps are therefore descent-flat: apply base change to a defining filtered presentation and take homotopy groups.

Lemma 2.2. *Adams-type maps of commutative ring spectra are stable under pushout.*

Proof. Suppose $A \rightarrow B$ is Adams-type and $A \rightarrow A'$ is commutative. Set $B' = A' \otimes_A B$. If P_α is an Adams presentation of B , then $A' \otimes_A P_\alpha$ is a perfect presentation of B' and

$$B' \otimes_{A'} (A' \otimes_A P_\alpha) \simeq B' \otimes_B (B \otimes_A P_\alpha).$$

The right side is homotopy-projective by Lemma 2.1. This is the commutative case of [BLP, Corollary 4]. $\square$

2.1 Differential graded models

Differentials lower degree by one. For a differential graded algebra C , we write $\mathcal{H}(C)$ for its associated associative algebra spectrum. For an ordinary ring this is the usual Eilenberg–Mac Lane spectrum HC . The equivalences of algebra and module homotopy theories in [Shi, Theorem 1.1 and Corollary 2.15] allow us to compute the derived tensors used below in unbounded differential graded modules. We distinguish a graded free module from a semifree differential graded module: in a semifree filtration the differential of each new generator lies in preceding stages.

Proposition 2.3. *A given diagram of strict commutative differential graded algebras over an ordinary commutative ring k admits a compatible realization as a diagram of commutative Hk -algebra spectra. Its underlying associative algebras and their module categories have the corresponding differential graded models. Consequently relative semifree resolutions compute the module tensors of that diagram.*

Proof. Fix a cofibrant E_∞ operad $\mathcal{O}$ in chain complexes together with its augmentation to the commutative operad. Restrict the given diagram along $\mathcal{O} \rightarrow \text{Com}$. Apply the comparison between E_∞ algebras in unbounded chain complexes and commutative Hk -algebras [RS, Corollary 8.3], with the compatible associative and module comparisons of [Shi, Corollary 2.15]. Equivalences of the resulting module infinity-categories preserve cofibers, retracts, and derived tensor products. In particular, a relative semifree resolution in the given diagram supplies the asserted tensor calculation. $\square$

We use Proposition 2.3 only in the indicated direction, starting with specified strict algebras and maps. Whenever odd squares are required to vanish in such an algebra, we state that convention explicitly. The proposition asserts no strictification of an arbitrary E_∞ algebra in positive characteristic.

2.2 Bar constructions

For an augmented commutative k -algebra A , set $\text{Bar}_k(A) = k \otimes_A k$. It is the suspension of A in the pointed category of augmented commutative k -algebras, and thus preserves colimits there. Its coproduct is induced by concatenation of based loops in $\text{CAlg}_k^{\text{op}}$; in the ordinary bar model it is the deconcatenation coproduct. These structures are natural in A . For $k = \mathbb{S}$ we abbreviate $\text{Bar}_{\mathbb{S}}(A)$ to $\text{Bar}(A)$.

Over $k = H\mathbb{F}_2$, homotopy groups turn a commutative Hopf k -algebra into a graded commutative Hopf algebra over $\mathbb{F}_2$. It need not be cocommutative. A homogeneous class z is *primitive* if $\Delta z = 1 \otimes z + z \otimes 1$. We call a coaugmented coalgebra *conilpotent* when each element of its augmentation coideal is killed by some iterated reduced coproduct. This condition concerns the coalgebra, not the connectivity of an underlying ring spectrum.

3 Descent-flat maps without Adams presentations

In this section we use the same notation for a DG algebra and its realization, via the comparisons recalled in Section 2. For DG modules we write $\otimes^{\mathbb{L}}$ when distinguishing a derived tensor product from a tensor product of chosen chain models.

Put

$$k = \mathbb{F}_2, \quad R = k[\varepsilon]/(\varepsilon^2),$$

where R is concentrated in degree zero. Consider the augmented associative DG R -algebra

$$A = R\langle y, x_0, x_1, \dots \rangle, \quad |y| = |x_i| = -1, \quad dy = 0, \quad dx_i = \varepsilon y x_{i+1}. \quad (3.1)$$

The coefficient ring R is central, and the augmentation $f : A \rightarrow R$ kills y and all x_i . The differential squares to zero because $d^2 x_i = \varepsilon^2 y^2 x_{i+2} = 0$.

Theorem 3.1. *The map $f : A \rightarrow R$ of (3.1) has the following properties.*

- (i) *Its derived self-tensor product has homology concentrated in degree zero, where it is free on generators $e, u, v_0, v_1, \dots$:*

$$H_*(R \otimes_A^{\mathbb{L}} R) \cong Re \oplus Ru \oplus \bigoplus_{i \geq 0} Rv_i.$$

- (ii) *The map f is not of Adams type.*

- (iii) *If $C = H_0(R \otimes_A^{\mathbb{L}} R)$ denotes its cooperation coalgebra, then*

$$H : \text{Mod}_A \longrightarrow \text{Comod}_C^{\text{gr}}, \quad H(X) = H_*(R \otimes_A^{\mathbb{L}} X),$$

is adapted in the sense of [PP, Definition 2.19].

In particular, a descent-flat map need not be of Adams type, even when its comodule-valued homology theory is adapted.

The obstruction to an Adams presentation is a natural square-zero operation on $H(X)$. At a projective stage, a single relation for this operation forces lifts of infinitely many independent classes; finite generation then gives a contradiction. We isolate this obstruction and the injective comparison needed for adaptedness before constructing the resolution that supplies both.

3.1 A finite projective obstruction

Lemma 3.2. *Let Λ be one of*

$$R, \quad R[t, t^{-1}], \quad R[v, v^{-1}],$$

where $|t| = 0$ and $|v| = 2$. Let the index set I be either $\mathbb{Z}_{\geq 0}$ or $\mathbb{Z}$, and put

$$N = \bigoplus_{i \in I} \Lambda z_i, \quad |z_i| = 0, \quad T_N(z_i) = \varepsilon z_{i+1}.$$

Then (N, T_N) is not a filtered colimit of graded Λ -modules equipped with a degree-zero square-zero operator whose underlying graded modules are finitely generated projective. The same conclusion holds for any operator module containing (N, T_N) as a direct summand.

Proof. Every projective graded Λ -module V satisfies

$$\text{ann}_V(\varepsilon) = \varepsilon V. \tag{3.2}$$

This is immediate for a free module from the description of Λ , and passes to direct summands.

Suppose that the stated filtered presentation exists. Lift z_0 to an element at one stage, then pass to a later stage where its relation $\varepsilon T_N z_0 = 0$ holds. Only this element and this one relation need to be realized at a common stage. Thus there are a finitely generated projective V , a square-zero operator T on V , an operator-preserving map $g : V \rightarrow N$, and an element $w_0 \in V_0$ such that

$$g(w_0) = z_0, \quad \varepsilon T w_0 = 0.$$

By (3.2), choose w_1 with $T w_0 = \varepsilon w_1$. Then $\varepsilon T w_1 = T^2 w_0 = 0$. Repeating this argument inside the same module V gives

$$T w_i = \varepsilon w_{i+1} \quad (i \geq 0).$$

Let $s : N \rightarrow N$ be the Λ -linear shift $s(z_i) = z_{i+1}$. Applying g gives

$$\varepsilon g(w_{i+1}) = \varepsilon s g(w_i).$$

Since (3.2) also holds for N , reduction modulo ε yields

$$\overline{g(w_i)} = z_i \quad \text{in } N/\varepsilon N \quad (i \geq 0).$$

The image of the finitely generated Λ/ε -module $V/\varepsilon V$ is supported on the union of the supports of the images of finitely many generators. This finite set cannot contain all the basis vectors (z_i) , a contradiction. For a larger target having N as an operator direct summand, lift z_0 in the larger target and then compose with its projection to N . The same argument applies. $\square$

Lemma 3.3. *Let B be an E_1 -ring and let Y be a perfect B -module. If $\pi_* Y$ is projective over $\pi_* B$, then it is finitely generated projective.*

Proof. The projectivity criterion in the proof of [BLP, Lemma 3] identifies Y as a retract of a coproduct of suspensions of B . Since Y is compact, its inclusion into that coproduct factors through a finite subcoproduct. Composing with the retraction exhibits Y as a retract of this finite subcoproduct. Applying π_* proves the assertion. $\square$

3.2 The comparison for injective comodules

For a graded coring Γ over a graded ring B_* , we use left comodules, with structure map $M \rightarrow \Gamma \otimes_{B_*} M$. Its counit will be denoted ϵ_Γ , to distinguish it from the element $\epsilon \in R$.

A graded homology theory H with values in an abelian category with enough injectives is *adapted* if, for each injective object J , the functor $\text{Hom}(H(-), J)$ is represented by an object E_J and the comparison $H(E_J) \rightarrow J$ corresponding to id_{E_J} is an isomorphism [PP, Definitions 2.11 and 2.19]. Thus adaptedness requires the comparison furnished by the representation to be an isomorphism.

Lemma 3.4. *Let $A \rightarrow B$ be a map of E_1 -rings, and suppose that*

$$\Gamma = \pi_*(B \otimes_A B)$$

is free as a graded right $B_ = \pi_* B$ -module. Then the canonical cooperations define a graded B_* -coring, and*

$$H(X) = \pi_*(B \otimes_A X)$$

with its canonical left Γ -coaction is an adapted homology theory.

Proof. Right freeness will supply the coring and reduce the injective comparison to cofree injectives. For those objects, the two adjunctions below determine both the representative and its comparison map.

The coring and its injectives. Write $F = B \otimes_A -$ and let G be restriction of scalars. As a right B -module, $B \otimes_A B$ is equivalent to a coproduct of suspensions of B : choose classes representing a right B_* -basis and take the resulting map from that coproduct. It is an equivalence because it induces an isomorphism on homotopy groups. Consequently, for every left B -module Y , the canonical map is an isomorphism

$$\Gamma \otimes_{B_*} \pi_* Y \xrightarrow{\cong} \pi_*(FGY). \quad (3.3)$$

The unit and counit of $F \dashv G$, together with (3.3), give the coring structure on Γ and the coaction on $H(X)$. Their identities follow from the comonad identities for FG .

Right flatness of Γ implies that the forgetful functor

$$U : \text{Comod}_\Gamma^{\text{gr}} \longrightarrow \text{Mod}_{B_*}^{\text{gr}}$$

is exact. Indeed, kernels and cokernels of comodule maps inherit their coactions from the corresponding module kernels and cokernels, using exactness of $\Gamma \otimes_{B_*} -$. Its right adjoint is the cofree comodule functor $J \mapsto \Gamma \otimes_{B_*} J$, which therefore preserves injectives. Every comodule M embeds in a cofree injective: choose an embedding $UM \rightarrow J$ into an injective graded module and compose

$$M \longrightarrow \Gamma \otimes_{B_*} UM \longrightarrow \Gamma \otimes_{B_*} J.$$

The first map is split injective after forgetting the coaction by the counit, and the second is injective by flatness. In particular, there are enough injectives, and every injective comodule is a retract of a cofree injective.

Representing cofree injectives. Let J be an injective graded B_* -module. It has a realization $I(J) \in \text{Mod}_B$ with $\pi_* I(J) = J$ and natural isomorphisms

$$[Y, I(J)]_B \cong \text{Hom}_{B_*}^{\text{gr}}(\pi_* Y, J);$$

this is [HL2, Proposition 1.2(1),(4)], applied to left modules, or equivalently to right modules over the opposite ring. Put $E_J = GI(J)$. Adjunction gives

$$\begin{aligned} [X, E_J]_A &\cong \mathrm{Hom}_{B_*}^{\mathrm{gr}}(H(X), J) \\ &\cong \mathrm{Hom}_{\Gamma\text{-comod}}^{\mathrm{gr}}(H(X), \Gamma \otimes_{B_*} J). \end{aligned} \quad (3.4)$$

Moreover, (3.3) identifies $H(E_J)$ with $\Gamma \otimes_{B_*} J$, carrying the cofree coaction $\Delta_\Gamma \otimes 1$.

The comparison map. The required map is the image of the identity under (3.4). Under $F \dashv G$, the identity of E_J corresponds to the counit $FGI(J) \rightarrow I(J)$. On homotopy groups this is $\epsilon_\Gamma \otimes 1_J$. Under the cofree adjunction, its corresponding comodule map is therefore

$$\theta_J = (1_\Gamma \otimes \epsilon_\Gamma \otimes 1_J)(\Delta_\Gamma \otimes 1_J) = 1_{\Gamma \otimes_{B_*} J}. \quad (3.5)$$

Thus the comparison map is an isomorphism, as required by [PP, Definitions 2.11 and 2.19].

Finally, let an injective comodule be the image of an idempotent on a cofree injective. The representation (3.4) and (3.5) lift this idempotent to an idempotent on E_J ; compatibility with composition follows from the natural representation. Splitting it in Mod_A gives a representing object whose comparison map is the corresponding retract of θ_J . This proves adaptedness for every injective comodule. $\square$

3.3 The semifree resolution and its operation

Proof of Theorem 3.1. The resolution below gives free cooperations and a natural operation. Freeness proves descent-flatness and permits the use of Lemma 3.4; the operation will exclude an Adams presentation by Lemma 3.2.

The resolution and the cooperation module. Let $I_A = \ker(A \rightarrow R)$. Decomposition of words by their first letter gives a decomposition of graded right A -modules

$$I_A = yA \oplus \bigoplus_{i \geq 0} x_i A.$$

The cone of $I_A \hookrightarrow A$ is the right DG module

$$\begin{aligned} P &= eA \oplus uA \oplus \bigoplus_{i \geq 0} v_i A, & |e| &= |u| = |v_i| = 0, \\ de &= 0, & du &= ey, & dv_i &= ex_i + \varepsilon u x_{i+1}. \end{aligned} \quad (3.6)$$

The second term in dv_i is the suspended differential of $x_i \in I_A$. In particular, the short exact sequence $0 \rightarrow I_A \rightarrow A \rightarrow R \rightarrow 0$ proves that $P \rightarrow R$ is a quasi-isomorphism. It is a semifree resolution: the filtration

$$0 \subset eA \subset eA \oplus uA \subset P$$

has free DG quotients. The differential of every v_i involves only the earlier generators e, u ; the later algebra generator x_{i+1} appears as a right coefficient. This is the semifree filtration needed to compute derived base change.

Tensoring (3.6) with R kills every displayed differential and gives

$$R \otimes_A^{\mathbb{L}} R \simeq P \otimes_A R = C = Re \oplus Ru \oplus \bigoplus_{i \geq 0} Rv_i \quad (3.7)$$

with zero differential. Both R -actions are the central coefficient action. This proves the first assertion and descent-flatness.

The operation and the Adams obstruction. Define a left R -linear, right A -linear map $S : P \rightarrow P$ by

$$S(e) = 0, \quad S(u) = e, \quad S(v_i) = \varepsilon v_{i+1}. \quad (3.8)$$

It has degree zero, $S^2 = 0$, and is a chain map. The only nontrivial check on generators is

$$dS(v_i) = \varepsilon e x_{i+1} = S(e x_i + \varepsilon u x_{i+1}) = S(dv_i).$$

Since P is semifree, $P \otimes_A -$ computes the derived base-change functor on all left DG A -modules. Thus $S \otimes 1$ induces a natural square-zero operation T on $H_*(R \otimes_A^{\mathbb{L}} X)$, compatible with filtered colimits. On (3.7) it gives the operator decomposition

$$(C, T) = (Re \oplus Ru, Tu = e, Te = 0) \oplus \left(\bigoplus_{i \geq 0} Rv_i, Tv_i = \varepsilon v_{i+1} \right).$$

Suppose that $R \simeq \operatorname{colim}_{\alpha} Q_{\alpha}$ were an Adams presentation over A . Base change preserves perfect objects, so Lemma 3.3 implies that $H_*(R \otimes_A^{\mathbb{L}} Q_{\alpha})$ is finitely generated projective as a graded R -module. In particular its degree-zero part is finitely generated projective. Applying the degree-zero homology and the natural operation gives

$$(C, T) \cong \operatorname{colim}_{\alpha} (H_0(R \otimes_A^{\mathbb{L}} Q_{\alpha}), T).$$

This contradicts Lemma 3.2 and proves that f is not of Adams type.

The cooperation coalgebra and adaptedness. The cooperation coalgebra in (3.7) has coproduct

$$\begin{aligned} \Delta e &= e \otimes e, \\ \Delta u &= e \otimes u + u \otimes e, \\ \Delta v_i &= e \otimes v_i + v_i \otimes e + \varepsilon u \otimes v_{i+1}. \end{aligned} \quad (3.9)$$

The counit sends e to 1 and all other generators to zero. To identify these formulas with the canonical cooperations, use the same formulas to define a chain map $\lambda : P \rightarrow C \otimes_R P$. The chain-map condition for v_i follows from

$$d\lambda(v_i) = e \otimes e x_i + \varepsilon e \otimes u x_{i+1} + \varepsilon u \otimes e x_{i+1} = \lambda(dv_i).$$

Its composite with $C \otimes_R P \rightarrow C$ is the canonical map $P \rightarrow P \otimes_A R$. The target map is a quasi-isomorphism because C is R -free, so λ represents the derived unit map on the right A -module R . Tensoring with R gives exactly the canonical coproduct (3.9). Its coassociativity can also be checked directly; the extra iterated term is divisible by ε^2 .

Finally, C is right R -free, so Lemma 3.4 applies. In particular its injective comparisons are the maps (3.5), proving the third assertion. $\square$

3.4 A finite algebraic presentation

The countably many generators in (3.1) can be encoded by conjugation by one invertible degree-zero generator.

Proposition 3.5. *Let $B_{\text{Laur}} = R[t, t^{-1}]$, concentrated in degree zero, and define the augmented DG algebra*

$$\begin{aligned} A_{\text{Laur}} &= R\langle t, t^{-1}, y, x \rangle / (tt^{-1} = t^{-1}t = 1, ty = yt), \\ |t| &= 0, \quad |y| = |x| = -1, \quad dt = dy = 0, \quad dx = \varepsilon y t x t^{-1}. \end{aligned} \quad (3.10)$$

The map $A_{\text{Laur}} \rightarrow B_{\text{Laur}}$ that kills x, y is descent-flat but not of Adams type. Its cooperation homology is concentrated in degree zero and is free on both sides over B_{Laur} . Its graded-comodule-valued homology theory is adapted.

Proof. Put $x_i = t^i x t^{-i}$ for $i \in \mathbb{Z}$, and let

$$D = R\langle y, x_i \mid i \in \mathbb{Z} \rangle, \quad dy = 0, \quad dx_i = \varepsilon y x_{i+1}.$$

The automorphism $\sigma(y) = y, \sigma(x_i) = x_{i+1}$ identifies A_{Laur} with the skew Laurent extension $D \rtimes_{\sigma} \mathbb{Z}$, with relation $ta = \sigma(a)t$. Indeed, the maps sending x_i to $t^i x t^{-i}$ in one direction, and x to x_0 in the other, are inverse on the generators.

The construction (3.6), now indexed by $\mathbb{Z}$, gives a right D -semifree resolution $P_D \rightarrow R$. Since $A_{\text{Laur}} = \bigoplus_{n \in \mathbb{Z}} D t^n$ as a left DG D -module,

$$P = P_D \otimes_D A_{\text{Laur}} \longrightarrow B_{\text{Laur}}$$

is a right semifree resolution. Give it the left B_{Laur} -action

$$t \cdot e = et, \quad t \cdot u = ut, \quad t \cdot v_i = v_{i+1}t. \quad (3.11)$$

The inverse action is defined by shifting i backwards and multiplying by t^{-1} . These actions commute with the differential: for example,

$$d(t \cdot v_i) = (ex_{i+1} + \varepsilon u x_{i+2})t = t \cdot (ex_i + \varepsilon u x_{i+1}).$$

Thus P is a $B_{\text{Laur}}\text{--}A_{\text{Laur}}$ bimodule resolution.

After tensoring with B_{Laur} , the differential vanishes and the right cooperation module is

$$C_{\text{Laur}} = eB_{\text{Laur}} \oplus uB_{\text{Laur}} \oplus \bigoplus_{i \in \mathbb{Z}} v_i B_{\text{Laur}}.$$

It is also left free on e, u, v_i . For the last summands, (3.11) sends the left-basis monomial $t^m \cdot v_i$ to the right-basis monomial $v_{i+m}t^m$; the change of indices $(i, m) \mapsto (i + m, m)$ is bijective. The two scalar actions need not coincide, so these cooperations form a coring rather than a central coalgebra over B_{Laur} .

The map $S(e) = 0, S(u) = e, S(v_i) = \varepsilon v_{i+1}$ is again a right-linear chain map with square zero. It also commutes with the left t -action, since

$$S(t \cdot v_i) = \varepsilon v_{i+2}t = t \cdot S(v_i).$$

As a left operator module, C_{Laur} has the direct summand $\bigoplus_i B_{\text{Laur}} v_i$ with $T(v_i) = \varepsilon v_{i+1}$. These are independent left basis vectors; the formula $t \cdot v_i = v_{i+1}t$ does not identify them by a left scalar multiple, since its final t is a right action. Lemmas 3.2 and 3.3 therefore exclude an Adams presentation exactly as in Theorem 3.1. Left freeness proves descent-flatness, and right freeness allows Lemma 3.4 to prove adaptedness. $\square$

Remark 3.6. The presentation (3.10) has finitely many algebra generators, relations, and differential formulas. This assertion concerns the DG algebra presentation; it does not assert compactness of its ring spectrum or perfection of B_{Laur} over A_{Laur} .

3.5 A connective source with a periodic target

Proposition 3.7. *Let A be the DG algebra (3.1), and put*

$$L = R[v, v^{-1}], \quad |v| = 2, \quad A_{\text{per}} = A \otimes_R L.$$

There is a connective E_1 -ring A_{cn} and a map $A_{\text{cn}} \rightarrow L$ which is descent-flat but not of Adams type. Its cooperation homology is free on both sides over L , and its graded-comodule-valued homology theory is adapted.

Proof. Use the following DG subalgebra of A_{per} :

$$(A_{\text{cn}})_n = \begin{cases} (A_{\text{per}})_n, & n > 0, \\ \ker(d : (A_{\text{per}})_0 \rightarrow (A_{\text{per}})_{-1}), & n = 0, \\ 0, & n < 0. \end{cases}$$

It is closed under multiplication and differential, and its inclusion realizes the connective cover of A_{per} . Restricting the augmentation gives $A_{\text{cn}} \rightarrow L$.

The element v is a central cycle in A_{cn} . Every homogeneous element a of A_{per} satisfies $v^m a \in A_{\text{cn}}$ for sufficiently large m , since one can arrange $|v^m a| > 0$. Hence ordinary central localization gives

$$A_{\text{cn}}[v^{-1}] = A_{\text{per}}. \quad (3.12)$$

It also computes derived localization: it is extension of scalars along the flat graded localization $R[v] \rightarrow R[v, v^{-1}]$, or equivalently the telescope of multiplication by v . Since v acts invertibly on L ,

$$A_{\text{per}} \otimes_{A_{\text{cn}}}^{\mathbb{L}} L \simeq L, \quad L \otimes_{A_{\text{cn}}}^{\mathbb{L}} L \simeq L \otimes_{A_{\text{per}}}^{\mathbb{L}} L.$$

Tensoring the semifree resolution (3.6) over R with L computes the latter object. Thus

$$H_*(L \otimes_{A_{\text{cn}}}^{\mathbb{L}} L) \cong Le \oplus Lu \oplus \bigoplus_{i \geq 0} Lv_i, \quad (3.13)$$

free on both sides, with all basis vectors in degree zero. The operation (3.8) extends L -linearly and gives $T(v_i) = \varepsilon v_{i+1}$. Lemma 3.2, now for graded L -modules, together with Lemma 3.3, proves that $A_{\text{per}} \rightarrow L$ is not of Adams type. The periodic unit changes degrees but does not change the independent basis indices i used in that obstruction.

If $L \simeq \text{colim}_{\alpha} Q_{\alpha}$ were an Adams presentation over A_{cn} , extension of scalars to A_{per} would give a presentation of L by the perfect modules $A_{\text{per}} \otimes_{A_{\text{cn}}} Q_{\alpha}$. Their base changes satisfy

$$L \otimes_{A_{\text{per}}} (A_{\text{per}} \otimes_{A_{\text{cn}}} Q_{\alpha}) \simeq L \otimes_{A_{\text{cn}}} Q_{\alpha},$$

so their homotopy groups are projective over L . This contradicts the preceding paragraph. Thus $A_{\text{cn}} \rightarrow L$ is not of Adams type. Finally, (3.13) proves descent-flatness and verifies the right-freeness hypothesis of Lemma 3.4, which proves adaptedness. $\square$

Remark 3.8. The target in Proposition 3.7 is nonconnective. The map has no ring-spectrum section: a section would send the invertible class $v^{-1} \in \pi_{-2}L$ to zero in the connective source, contradicting $vv^{-1} = 1$. The propositions above concern E_1 -rings; they make no assertion that these noncommutative sources can be replaced by E_{∞} -rings.

4 Flat base change without perfect approximation

We first give a counterexample with discrete commutative source and target. Throughout this section complexes are homologically graded, so their differentials have degree -1 . Set

$$R = \mathbb{F}_2[\varepsilon]/(\varepsilon^2), \quad A = R[y], \quad B = R,$$

where y has degree zero, and let $A \rightarrow B$ send y to zero. Define a complex of A -modules by

$$M = \bigoplus_{i \geq 0} Am_i, \quad |m_i| = -i, \quad dm_i = \varepsilon y m_{i+1}. \quad (4.1)$$

The relation $\varepsilon^2 = 0$ gives $d^2 = 0$. We use the same notation for this complex and the corresponding HA -module.

Theorem 4.1. *The map $HA \rightarrow HB$ is an Adams-type map of $\mathbb{E}_\infty$ -rings, and*

$$\pi_*(HB \otimes_{HA} M) \cong \bigoplus_{i \geq 0} Rz_i, \quad |z_i| = -i,$$

is free as a graded R -module. Nevertheless, M is not a filtered colimit of perfect HA -modules Q for which $\pi_(HB \otimes_{HA} Q)$ is projective over R .*

The obstruction retains a natural operation on base-change homology. A four-cell perfect module will record one element and one relation that cannot pass through a perfect module with projective base-change homology. Compactness forces precisely such a factorization in any filtered presentation, giving the contradiction.

4.1 A natural operation and a finite-support obstruction

For any A -complex X , write

$$B_*X = H_*(B \otimes_A^{\mathbb{L}} X).$$

All the calculations below take place in the unbounded derived category. The comparison with HA -modules, including the module categories, is supplied by [Shi, Theorem 1.1 and Corollary 2.15].

Lemma 4.2. *The functor $B_*(-)$ has a natural R -linear operation*

$$\beta_X : B_n X \longrightarrow B_{n-1} X, \quad \beta_X^2 = 0.$$

*For the complex M in (4.1), there is a graded R -basis $(z_i)_{i \geq 0}$ of B_*M satisfying*

$$\beta_M(z_i) = \varepsilon z_{i+1}. \tag{4.2}$$

Proof. Since y is a non-zero-divisor in $R[y]$, the complex

$$P = eA \oplus uA, \quad |e| = 0, \quad |u| = 1, \quad de = 0, \quad du = ey \tag{4.3}$$

is a right A -module resolution of B . It also has its evident left R -action. It is a bounded complex of free right A -modules and hence is K -flat. Consequently $P \otimes_A X$ computes $B \otimes_A^{\mathbb{L}} X$ for every unbounded complex X .

The left R -linear, right A -linear map of degree -1

$$S(e) = 0, \quad S(u) = e$$

satisfies $dS + Sd = 0$ and $S^2 = 0$. Thus $S \otimes 1$ induces β_X . Because $P \otimes_A -$ preserves quasi-isomorphisms, this operation is natural for all morphisms in the derived category.

For $X = M$, the complex $P \otimes_A M$ is the mapping cone of $y : M \rightarrow M$. Multiplication by y is injective in every degree, so the projection

$$P \otimes_A M \longrightarrow M/yM$$

is a quasi-isomorphism. This follows from the short exact sequence of complexes for multiplication by y and does not require a lower bound on M . The quotient has zero differential and is free on the images z_i of the m_i .

A cycle representing z_i is

$$c_i = e \otimes m_i + \varepsilon u \otimes m_{i+1}.$$

Indeed, the two terms involving εy in its differential cancel, and the remaining term has coefficient ε^2 . Finally,

$$S(c_i) = \varepsilon e \otimes m_{i+1} = \varepsilon c_{i+1},$$

which proves (4.2). □

We record the algebraic obstruction in a form which also applies to the periodic example below.

Lemma 4.3. *Let K be either R or $R[v, v^{-1}]$, where $|v| = 2$. Put*

$$N_K = \bigoplus_{i \geq 0} Kz_i, \quad |z_i| = -i, \quad T_N(z_i) = \varepsilon z_{i+1}.$$

There is no finitely generated projective graded K -module V , equipped with a K -linear operation T_V of degree -1 and square zero, admitting an operation-preserving graded map $g : V \rightarrow N_K$ and an element $w_0 \in V_0$ such that

$$g(w_0) = z_0, \quad \varepsilon T_V w_0 = 0.$$

In particular, N_K , with its operation, is not a filtered colimit of such finitely generated projective modules.

Proof. Every projective graded K -module satisfies

$$\text{ann}_V(\varepsilon) = \varepsilon V :$$

this holds for graded free modules and passes to retracts. Starting from w_0 , choose homogeneous elements recursively so that

$$T_V w_i = \varepsilon w_{i+1}.$$

The recursion continues because $\varepsilon T_V w_{i+1} = T_V^2 w_i = 0$. Thus one initial relation produces the whole sequence inside the same finitely generated module.

Let $s : N_K \rightarrow N_K$ send z_i to z_{i+1} . Since $T_N = \varepsilon s$, naturality gives

$$\varepsilon g(w_{i+1}) = \varepsilon s g(w_i).$$

Using $\text{ann}_{N_K}(\varepsilon) = \varepsilon N_K$, reduction modulo ε therefore yields

$$\overline{g(w_i)} = z_i \quad (i \geq 0).$$

The image of the finitely generated $K/(\varepsilon)$ -module $V/\varepsilon V$ in $\bigoplus_{i \geq 0} (K/(\varepsilon))z_i$ is supported on a finite set of basis indices: take the union of the supports of finitely many generators. It cannot contain every z_i .

For the last assertion, a filtered presentation would allow z_0 to be lifted to some stage. The relation $\varepsilon T_N z_0 = 0$ holds at a later stage, giving exactly the prohibited pair (g, w_0) . Only one element and one vanishing relation must occur at that stage, so this argument applies to arbitrary transition maps. $\square$

Proof of Theorem 4.1. The two-cell resolution (4.3) shows that B is perfect over A and that

$$B \otimes_A^{\mathbb{L}} B \cong B \oplus \Sigma B.$$

The constant diagram with value B is consequently an Adams-type presentation of B over A . Lemma 4.2 gives the asserted free base-change homology of M .

It remains to exclude the perfect approximations. We realize the element and relation of Lemma 4.3 on a finite complex, so that compactness can detect them at a single stage of any proposed filtered presentation.

A finite witness for the operator relation. Consider the perfect A -complex

$$\begin{aligned} L_0 &= Aa \oplus Ac, & L_{-1} &= Ab \oplus Ad, \\ \partial a &= yb, & \partial c &= \varepsilon b + yd, & \partial b &= \partial d = 0. \end{aligned} \tag{4.4}$$

There is a chain map $g : L \rightarrow M$ given by

$$g(a) = m_0, \quad g(b) = \varepsilon m_1, \quad g(c) = g(d) = 0.$$

In $P \otimes_A L$, the degree-zero cycle $w = e \otimes a + u \otimes b$ satisfies

$$\beta_L[w] = [e \otimes b], \quad d(e \otimes c + u \otimes d) = \varepsilon e \otimes b.$$

Hence

$$\varepsilon \beta_L[w] = 0, \quad g_*[w] = z_0. \tag{4.5}$$

The cells a, b provide the lift of z_0 , while c, d make the required relation hold in homology. Both properties are therefore preserved by any factorization of g .

Factorization through a good perfect module. Suppose that g factored in the derived category through a perfect A -module Q with projective graded base-change homology $V = B_*Q$. The B -module $B \otimes_A^{\mathbb{L}} Q$ is perfect. A module with projective homotopy groups is a retract of a sum of shifts of the ring, as observed in the proof of [BLP, Lemma 3]. Since this module is compact, the map into that sum factors through a finite subsum. Thus V is finitely generated projective. The image $w_0 \in V_0$ of $[w]$ satisfies (4.5), and the induced map $V \rightarrow B_*M$ commutes with β . Lemma 4.3, with $K = R$, gives a contradiction.

Finally, if M were a filtered colimit of the required perfect modules Q_α , compactness of L would imply that the class of $g : L \rightarrow M$ factors through some Q_α . Perfect modules are precisely the compact objects of the module category [HA, Proposition 7.2.4.2]. This proves the claim. $\square$

Remark 4.4. The rings in Theorem 4.1 are discrete commutative rings. The unboundedness is entirely in M . Although its underlying graded A -module is free, no K -flatness of M was used: the derived tensor product was computed using the right resolution P .

4.2 A connective source and a connective module

Let $\mathcal{H}$ denote the realization of a specified strict commutative DG model, with the compatibility described in Proposition 2.3. In particular, the following models determine an actual $\mathbb{E}_\infty$ -map:

$$A_+ = \mathcal{H}(R[y, v]), \quad B_{\text{per}} = \mathcal{H}(R[v, v^{-1}]), \quad |y| = 0, \quad |v| = 2, \quad y \mapsto 0.$$

The differentials of these algebras are zero.

Theorem 4.5. *The map $A_+ \rightarrow B_{\text{per}}$ is Adams-type. There is a connective A_+ -module M_+ such that $\pi_*(B_{\text{per}} \otimes_{A_+} M_+)$ is free over π_*B_{per} , but M_+ is not a filtered colimit of perfect A_+ -modules with projective B_{per} -base-change homotopy.*

Proof. The perfect A_+ -module

$$K = \text{cofib}(y : A_+ \rightarrow A_+)$$

is represented by $R[v]$. Multiplication by v gives a sequence $\Sigma^{-2n}K \rightarrow \Sigma^{-2n-2}K$ whose colimit is B_{per} . Moreover,

$$B_{\text{per}} \otimes_{A_+} \Sigma^{-2n}K \simeq \Sigma^{-2n}(B_{\text{per}} \oplus \Sigma B_{\text{per}}).$$

This is an Adams-type presentation.

Over $D = R[y, v, v^{-1}]$, form the complex

$$\widetilde{M} = \bigoplus_{i \geq 0} Dm_i, \quad |m_i| = -i, \quad dm_i = \varepsilon y m_{i+1},$$

and set $M_+ = \tau_{\geq 0} \widetilde{M}$ as an A_+ -module. The connective cover is an A_+ -module because A_+ is connective. In the DG model it retains the positive chain groups, takes cycles in degree zero, and vanishes in negative degrees. Every homogeneous chain element of $\widetilde{M}$, after multiplication by a sufficiently large power of v , lies in a strictly positive degree. It follows that

$$M_+[v^{-1}] \simeq \widetilde{M}.$$

The localization is computed by the telescope of multiplication by v ; on complexes it is exact. Thus this is also the derived localization, and

$$B_{\text{per}} \otimes_{A_+} M_+ \simeq (R[v, v^{-1}]) \otimes_D^{\mathbb{L}} \widetilde{M}.$$

Use the right D -module resolution

$$eD \oplus uD, \quad |u| = 1, \quad du = ey.$$

The calculation of Lemma 4.2 now gives

$$\pi_*(B_{\text{per}} \otimes_{A_+} M_+) \cong \bigoplus_{i \geq 0} R[v, v^{-1}]z_i, \quad \beta z_i = \varepsilon z_{i+1}.$$

For an arbitrary A_+ -module the same operation is defined after derived localization at v , so it remains natural in the original module category.

If a filtered presentation as in the theorem existed, base change and homotopy groups would give a filtered presentation of this module with its operation. Each stage would have finitely generated projective graded homotopy, by perfectness and the compactness argument used above. This contradicts Lemma 4.3 with $K = R[v, v^{-1}]$. $\square$

Remark 4.6. Here B_{per} is not connective. Also, the module M_+ is not almost perfect. Indeed,

$$H_*(\widetilde{M}) = \varepsilon D m_0 \oplus \bigoplus_{i \geq 1} (\varepsilon D / \varepsilon y D) m_i,$$

and hence, as an $R[y]$ -module,

$$\pi_0 M_+ \cong \mathbb{F}_2[y] \oplus \bigoplus_{j \geq 1} \mathbb{F}_2.$$

In the second summand both y and ε act trivially. This module is not finitely generated. The example therefore does not impose simultaneous connectivity on the source, target, and module, and it does not fall under Theorem 11.1.

5 A transfinite obstruction for associative ring spectra

Fix a prime p and put

$$A = H\mathbb{Z} \times H\mathbb{F}_p, \quad N = e_z N e_p = H\mathbb{F}_p, \quad B = A \oplus N = H \begin{pmatrix} \mathbb{Z} & \mathbb{F}_p \\ 0 & \mathbb{F}_p \end{pmatrix}. \quad (5.1)$$

Here e_z and e_p are the two idempotents of A . The left $H\mathbb{Z}$ -action on N is reduction modulo p , the right $H\mathbb{F}_p$ -action is the regular action, and the other three corners vanish. The ring B is the canonical square-zero extension with this bimodule. Write $j : A \rightarrow B$ for the diagonal inclusion and $\rho : B \rightarrow A$ for the diagonal projection.

Theorem 5.1. *The map j in (5.1) is not a continuous transfinite composite of Adams-type maps of E_1 -ring spectra. More strongly, it is not a continuous transfinite composite of descent-flat maps. This holds for every small ordinal length, without any connectivity or boundedness assumption on the intermediate rings.*

We use the left-flat convention of [BLP, Definition 1]. The obstruction is the vanishing of mapping spectra into N from all bimodules that are homotopy-flat on the left over A . At the initial ring A , this vanishing follows from the two characteristics at its vertices. To propagate it through a successor step, we first prove that tensoring with the cooperation bimodule preserves homotopy-flatness on the left over A , then use a relative module bar resolution. At limit stages, compatible actions over the earlier rings determine the mapping spectrum. Finally, at B the relative tensor $A \otimes_B A$ supplies a nonzero map from a shifted free left A -module, contradicting the propagated vanishing.

All tensors in this section are derived. The notation $\text{Bimod}(A, T)$ means left- A , right- T bimodules in spectra, with commuting coherent actions; it does not require a central A -algebra structure on T . A left A -module is *homotopy-flat* if its graded homotopy is flat over π_*A .

Lemma 5.2. *Let $T \rightarrow U$ be a map of E_1 -rings and set $\Gamma = U \otimes_T U$, with its two U -actions. If $T \rightarrow U$ is Adams-type, then for every (A, U) -bimodule F that is homotopy-flat as a left A -module, the bimodule $F \otimes_U \Gamma$ is again homotopy-flat as a left A -module. The same conclusion holds if $T \rightarrow U$ is merely descent-flat.*

Proof. We first prove the Adams-type assertion directly. Choose a filtered presentation $U \simeq \text{colim}_i P_i$ by perfect left T -modules such that $Q_i = U \otimes_T P_i$ has projective graded homotopy over π_*U . A module with projective graded homotopy is an actual retract of a coproduct of shifts of the ring: realize a free presentation of its homotopy groups, realize the corresponding idempotent on the free module, and split it in the module category. This is also the projectivity criterion used in [BLP, Lemma 3]. Hence $F \otimes_U Q_i$ is, as a left A -module, a retract of a coproduct of shifts of F . Since

$$\Gamma \simeq \text{colim}_i Q_i$$

as left U -modules, tensoring with F and taking homotopy proves the claim. Flat graded modules are closed under coproducts, retracts, shifts, and filtered colimits.

This argument uses only a left U -module presentation of Γ . No right U -linear refinement of that presentation is needed. The resulting tensor independently retains its right U -action from the actual bimodule Γ .

For the descent-flat assertion, apply the flat tensor criterion of [HL1, Proposition 2.3, (1) $\Leftrightarrow$ (5)] to U^{op} . It gives an isomorphism, natural in F and compatible with the left π_*A -action,

$$\pi_*(F \otimes_U \Gamma) \cong \pi_*F \otimes_{\pi_*U} \pi_*\Gamma. \quad (5.2)$$

The cited statement applies to arbitrary ring spectra and already given modules; it imposes no boundedness assumption. To see the remaining flatness directly, let X be any graded right π_*A -module. Associativity gives

$$X \otimes_{\pi_*A} (\pi_*F \otimes_{\pi_*U} \pi_*\Gamma) \cong (X \otimes_{\pi_*A} \pi_*F) \otimes_{\pi_*U} \pi_*\Gamma.$$

The first tensor functor is exact because F is left A -homotopy-flat, and its output retains a right π_*U -action. The second is exact because $\pi_*\Gamma$ is left π_*U -flat. Their composite is exact, proving the assertion by (5.2). $\square$

Consider now an E_1 -ring T equipped with compatible maps $A \rightarrow T \xrightarrow{\rho_T} A$ whose composite is the identity. Give N its right T -action through ρ_T , and define

$$\mathcal{P}(T) : \quad \text{Map}_{\text{Bimod}(A,T)}(F, N) \simeq 0 \quad \text{for every left-}A\text{-homotopy-flat } F. \quad (5.3)$$

The quantifier ranges over actual bimodules with all their coherent actions. Vanishing of the entire mapping spectrum allows us to take limits in both the bar resolution and the limit-ordinal argument, retaining all the coherence data of equivariant maps.

Lemma 5.3. *Let $T \rightarrow U$ be a map over and under A . If it is Adams-type or descent-flat, then $\mathcal{P}(T)$ implies $\mathcal{P}(U)$.*

Proof. Fix a left- A -homotopy-flat (A, U) -bimodule F . We will resolve F by bimodules induced from T , with sources homotopy-flat on the left over A to which $\mathcal{P}(T)$ applies. For the extension–restriction adjunction on right actions,

$$- \otimes_T U : \text{Bimod}(A, T) \rightleftarrows \text{Bimod}(A, U) : \text{Res},$$

the augmented bar resolution of F has terms

$$C_n = F \otimes_T U^{\otimes_T(n+1)} = D_n \otimes_T U, \quad D_n = F \otimes_T U^{\otimes_T n}. \quad (5.4)$$

In particular $D_0 = F$. Every D_n has a right U -action, and associativity identifies

$$D_n \simeq F \otimes_U \Gamma^{\otimes_U n}, \quad D_{n+1} \simeq D_n \otimes_U \Gamma, \quad \Gamma = U \otimes_T U. \quad (5.5)$$

The tensor with F uses the first, left U -action on Γ ; the last right action is retained. Lemma 5.2 therefore shows inductively that every D_n is homotopy-flat on the left over A . Thus each C_n is induced from a bimodule to which $\mathcal{P}(T)$ applies; it remains to recover F from these terms.

The realization of (5.4) is F as an actual (A, U) -bimodule. Indeed, after restricting to T the augmented simplicial object is split by the unit insertions of the adjunction; its augmentation uses the right U -action on F . Restriction preserves geometric realizations and is conservative, so this split resolution computes F before as well as after restriction. No convergence assertion about homotopy groups is involved.

The right T -action on N is the restriction of its right U -action. Consequently adjunction and $\mathcal{P}(T)$ give

$$\text{Map}_{\text{Bimod}(A,U)}(C_n, N) \simeq \text{Map}_{\text{Bimod}(A,T)}(D_n, N) \simeq 0$$

for every n . Mapping out of the realization turns it into a limit of zero spectra. Hence $\text{Map}_{\text{Bimod}(A,U)}(F, N) \simeq 0$, proving $\mathcal{P}(U)$. $\square$

Lemma 5.4. *Let β be a nonzero small limit ordinal and let $T_\beta \simeq \text{colim}_{\alpha < \beta} T_\alpha$ be a continuous diagram over and under A . If $\mathcal{P}(T_\alpha)$ holds for all $\alpha < \beta$, then $\mathcal{P}(T_\beta)$ holds.*

Proof. Fix actual (A, T_β) -bimodules F, N , with F homotopy-flat on the left over A . Restrict their right actions to all earlier stages. Their underlying left A -modules do not change.

An action of T_β^{op} on an object of Mod_A is an algebra map to its endomorphism algebra. The universal property of the algebra colimit identifies such actions with compatible actions of all T_α^{op} . Applying the same description in the arrow category of Mod_A identifies coherent equivariant maps. The indexing ordinal is nonempty and contractible, so a compatible system of underlying left A -modules is equivalent to a single such module. It follows that the restriction functors identify

$$\text{Bimod}(A, T_\beta) \simeq \lim_{\alpha < \beta} \text{Bimod}(A, T_\alpha).$$

Equivalently, one may use modules over $A \otimes T_\alpha^{\text{op}}$: these rings have colimit $A \otimes T_\beta^{\text{op}}$, since the diagram is filtered. Applying the mapping-space statement to all suspensions gives

$$\text{Map}_{\text{Bimod}(A, T_\beta)}(F, N) \simeq \lim_{\alpha < \beta} \text{Map}_{\text{Bimod}(A, T_\alpha)}(F, N). \quad (5.6)$$

Every spectrum on the right is zero. Its limit is therefore zero, proving $\mathcal{P}(T_\beta)$. In particular, this argument does not try to glue separately chosen nullhomotopies and does not leave a derived-limit ambiguity. $\square$

Proof of Theorem 5.1. We first establish $\mathcal{P}(A)$. If F is an A -bimodule homotopy-flat on the left, its corner $e_z F e_p$ is a retract of its left $H\mathbb{Z}$ part. Hence every group $\pi_n(e_z F e_p)$ is torsion-free over $\mathbb{Z}$. The right $H\mathbb{F}_p$ -action also makes these groups annihilated by p . The two integer actions restrict to the same scalar action of the sphere, so $e_z F e_p = 0$. The bimodule category for a product ring decomposes into its four corner categories. Since N is supported only in the (z, p) corner, this proves the vanishing of the full spectrum $\text{Map}_{\text{Bimod}(A, A)}(F, N)$.

Suppose that j admitted a continuous transfinite factorization $(T_\alpha)_{\alpha \leq \lambda}$ of the asserted kind. Composition with $T_\alpha \rightarrow T_\lambda = B \xrightarrow{\rho} A$ gives compatible retractions ρ_{T_α} . Thus Lemmas 5.3 and 5.4 apply throughout the diagram. Transfinite induction implies $\mathcal{P}(B)$.

To contradict this, give A its right B -action through ρ and take $F = \Sigma^{-1}A$. This is a shifted free left A -module. The corner supports give $N \otimes_A N = 0$ as an actual bimodule. The normalized relative bar for the canonical square-zero algebra $B = A \oplus N$ consequently has only lengths zero and one, giving

$$A \otimes_B A \simeq A \oplus \Sigma N \quad \text{in } \text{Bimod}(A, A). \quad (5.7)$$

One can also compute this by the two-term right B -module resolution $N \rightarrow B \rightarrow A$. Here $N \cong e_p B$ as a right B -module, so the first two terms are right B -projective. After tensoring with A they become $N \rightarrow A$ with the original left A -actions retained. This map is zero in the bimodule category, since its source is supported in the (z, p) corner and its target in the two diagonal corners. Its cofiber is precisely (5.7).

Extending the right action along $B \rightarrow A$ now gives

$$\begin{aligned} \text{Map}_{\text{Bimod}(A, B)}(\Sigma^{-1}A, N) &\simeq \text{Map}_{\text{Bimod}(A, A)}(\Sigma^{-1}(A \otimes_B A), N) \\ &\simeq \text{Map}_{\text{Bimod}(A, A)}(\Sigma^{-1}A \oplus N, N). \end{aligned}$$

Projection onto N is nonzero: its restriction to that summand is id_N . Thus $\mathcal{P}(B)$ fails. This proves both assertions, using respectively the direct Adams-type and descent-flat parts of Lemma 5.2. $\square$

Remark 5.5. The corner direction is essential. If instead $N^- = e_p N^- e_z = H\mathbb{F}_p$ and $B^- = A \oplus N^-$, then ${}_A B^- \simeq A \oplus A e_p$ is finite projective. The single perfect witness B^- has

$$B^- \otimes_A B^- \simeq B^- \oplus B^- e_p$$

as a left B^- -module. Thus $A \rightarrow B^-$ is itself Adams-type. This also checks the role of the left-flat convention in the theorem.

The target in (5.1) is noncommutative. Theorem 5.1 therefore answers the associative part of [BLP, Question (2)]; it makes no assertion about a factorization constrained to commutative intermediate rings.

6 Sphere-flat relative algebras and power operations

We begin the commutative factorization argument with the consequence of the Adams condition that will control power operations. The augmentation lets us turn projectivity after base change into a presentation by sphere retracts. All homology classes then lift from spheres, where an extended-square calculation will force the required operations to vanish. All augmentations in this section take values in the sphere.

Lemma 6.1. *Let $A \rightarrow B$ be an Adams-type map of augmented commutative sphere algebras, compatible with the augmentations, and put $C = B \otimes_A \mathbb{S}$. The underlying spectrum of C is a filtered colimit of retracts of finite coproducts of shifted spheres. In particular it is homotopy-flat over $\mathbb{S}$. Every element of $H_d(C; \mathbb{F}_2)$, for any integer d , is the Hurewicz image of a map $\Sigma^d \mathbb{S} \rightarrow C$, and every positive Steenrod homology operation on $H_*(C; \mathbb{F}_2)$ vanishes.*

Proof. Lemma 2.2 makes $\mathbb{S} \rightarrow C$ Adams-type. Choose its defining presentation $C \simeq \text{colim}_\alpha P_\alpha$. Thus P_α is a perfect spectrum and $C \otimes P_\alpha$ is homotopy-projective over C . The given augmentation supplies a map $C \rightarrow \mathbb{S}$. Base changing a free-retract presentation along this map gives

$$\mathbb{S} \otimes_C (C \otimes P_\alpha) \simeq P_\alpha$$

as a projective sphere module. Compactness and Lemma 2.1 make it a retract of a finite coproduct of shifted spheres.

For such a finite coproduct the Hurewicz map onto mod-two homology is surjective in every degree, and all positive Steenrod homology operations vanish. These properties pass to retracts and to filtered colimits. This proves the assertions about C . $\square$

The augmentation in this lemma is essential: it is the map used to recover P_α from $C \otimes P_\alpha$. No analogous assertion is being made for an arbitrary Adams-type unit $\mathbb{S} \rightarrow C$ without an augmentation.

For a commutative sphere algebra, use upper-index mod-two Dyer–Lashof operations with the convention

$$Q^i : H_d(-; \mathbb{F}_2) \longrightarrow H_{d+i}(-; \mathbb{F}_2), \quad Q^i z = 0 \ (i < d), \quad Q^d z = z^2.$$

Here d and i are integers.

Proposition 6.2. *If the underlying spectrum of a commutative sphere algebra C has the filtered presentation of Lemma 6.1, then*

$$Q^i H_*(C; \mathbb{F}_2) = 0 \quad \text{for every integer } i \neq 0.$$

The assertion does not include Q^0 .

Proof. The aim is to express each $Q^i z$, for $i \neq 0$, as the image of a positive Steenrod operation in an extended square. Its image in C must then vanish by Lemma 6.1. Take a homogeneous class z of degree d and choose its spherical lift. The commutative multiplication then supplies a map

$$D_2(\Sigma^d \mathbb{S}) \longrightarrow C, \quad D_2(X) = (X \otimes X)_{h\Sigma_2}.$$

For all integers d , this extended square is the Thom spectrum of the virtual bundle $d(1 \oplus \gamma)$ over $\mathbb{RP}^\infty$, where γ is the real sign line. This description and the upper-index convention for its classes are given in [Bak, Section 4.1, proof of Theorem 4.1].

Let $t \in H^1(\mathbb{RP}^\infty; \mathbb{F}_2)$, and let U be the Thom class of degree $2d$. The Thom and Cartan formulas give

$$Sq^r(t^j U) = \binom{d+j}{r} t^{j+r} U \quad (j, r \geq 0).$$

For negative d this uses the coefficient expansion of $(1+t)^d$ for the virtual bundle; it can equally be computed on each finite skeleton and then desuspended. Let e_j be dual to $t^j U$. Its degree is $2d+j$, its image in C is $Q^{d+j}z$, and

$$Sq_*^r(e_{j+r}) = \binom{d+j}{r} e_j. \quad (6.1)$$

Put $s = d+j \neq 0$. Write $s = 2^v u$, with u odd, and choose $r = 2^v > 0$. In $\mathbb{F}_2[[t]]$ the coefficient of t^{2^v} in $(1+t)^s = (1+t^{2^v})^u$ is one, also when $u < 0$. Thus the coefficient in (6.1) is one. Naturality and the vanishing of positive Steenrod operations on C force the image of e_j , namely $Q^s z$, to vanish. The indices $s < d$ vanish by instability, completing the proof. $\square$

After the strong symmetric monoidal base change $\mathbb{S} \rightarrow H\mathbb{F}_2$, these are exactly the relative operations on the homotopy groups of $H\mathbb{F}_2 \otimes C$. Indeed base change identifies the corresponding extended powers. This observation lets us use Proposition 6.2 inside the relative bar construction over $H\mathbb{F}_2$.

It remains to transfer the operation constraint from C to the kernel of a map on bar homology. An ordinary homology quotient is insufficient for this purpose, because the relative tensor product is derived. The next section supplies its filtered Hopf structure and identifies the coaction on its first homology page.

7 The Hopf structure on the relative bar filtration

Let k be a commutative ring spectrum, and let $A \rightarrow B$ be a map of augmented commutative k -algebras. In this section set

$$C = B \otimes_A k, \quad D = k \otimes_A k, \quad E = k \otimes_B k \simeq D \otimes_C k. \quad (7.1)$$

Thus D and E are the bar algebras of the source and target, and E is obtained from D by base change along $C \rightarrow k$. The presentation $E \simeq D \otimes_C k$ provides the filtration used below. The map $f : C \rightarrow D$ uses the augmentation $B \rightarrow k$, and $q : D \rightarrow E$ is the induced map. The equivalence includes the specified homotopy $qf \simeq \eta \varepsilon_C$. Let $F_s E$ denote the ordinary skeletal filtration of the two-sided module bar $B_\bullet(D, C, k)$.

Theorem 7.1. *The filtration $F_\bullet E$ admits a natural Hopf commutative algebra structure in filtered k -modules. Its coproduct takes values in the sum filtration:*

$$\Delta : F_\bullet E \longrightarrow F_\bullet E \otimes_{\text{Day}} F_\bullet E.$$

After forgetting filtration, this is the canonical Hopf structure on $\text{Bar}_k(B)$, and the map from $F_0 E = D$ is the canonical Hopf map $\text{Bar}_k(A) \rightarrow \text{Bar}_k(B)$. If $I = \text{fib}(C \rightarrow k)$, then

$$\text{gr}_s F_\bullet E \simeq \Sigma^s (D \otimes_k I^{\otimes k^s}).$$

No connectivity or boundedness assumption is imposed.

There are two identifications to establish. A homotopy stabilizer will supply a filtered group object, whose underlying filtration must be identified with the ordinary bar skeletons and their attaching maps. We then compute the zero-column component of its coproduct in Proposition 7.3. This second identification determines the canonical coinvariants used in the descent argument of Section 8.

7.1 The unit filtration and homotopy stabilizers

Write $\text{Fil}(\text{Mod}_k) = \text{Fun}(\mathbb{N}_{\leq}, \text{Mod}_k)$, with increasing filtrations and Day convolution for addition on $\mathbb{N}$. For two filtrations F, G , this tensor product is given by

$$(F \otimes_{\text{Day}} G)_s \simeq \text{colim}_{a+b \leq s} (F_a \otimes_k G_b).$$

Thus a coproduct with this target splits the available filtration degree between its two factors. For $a \geq 0$, let $\text{step}_a(M)$ be zero below a and equal to M from a onward. Then

$$\text{step}_a(M) \otimes_{\text{Day}} \text{step}_b(N) \simeq \text{step}_{a+b}(M \otimes_k N). \quad (7.2)$$

In particular step_0 is the constant-filtration, strong symmetric monoidal functor, and the monoidal unit is $\text{step}_0(k)$. Taking the colimit of a filtration is also strong symmetric monoidal.

Give C the *unit filtration*

$$\tilde{C}_0 = k, \quad \tilde{C}_s = C \quad (s \geq 1),$$

whose first structure map is the unit. To give it a commutative algebra structure, apply step_1 to the nonunital E_∞ augmentation ideal of C , using $\text{step}_n \rightarrow \text{step}_1$ for its n -ary products, and adjoin the filtered unit. All operadic compatibilities are inherited from that ideal. On underlying filtered modules the augmentation gives a splitting

$$\tilde{C} \simeq \text{step}_0(k) \oplus \text{step}_1(I). \quad (7.3)$$

The multiplication of two ideal factors is still the original multiplication of C , now with target in filtration two.

Lemma 7.2. *In an infinity-category with finite limits, let a group object G act on X , and choose a point $x : * \rightarrow X$. The homotopy fiber $G \times_X *$ of the orbit map is naturally a group object. Applied in $\text{CAlg}(\mathcal{V})^{\text{op}}$, where $\mathcal{V}$ is Mod_k or $\text{Fil}(\text{Mod}_k)$, this gives a Hopf commutative algebra*

$$D \otimes_C k$$

from a Hopf commutative algebra D , a right comodule algebra $\delta : C \rightarrow C \otimes D$, and an augmentation of C ; the map $C \rightarrow D$ is $(\varepsilon_C \otimes \text{id})\delta$.

Proof. Form the action groupoid with n -simplices $X \times G^n$. Pull it back along the chosen point at all $n+1$ vertices. The resulting simplicial object has object space $*$ and arrow space

$$* \times_X (X \times G) \times_X * \simeq G \times_X *.$$

The Segal maps and the inverse identities are preserved by these finite homotopy pullbacks. This supplies all group coherences, not just a multiplication on the arrow space. The dual statement follows because finite limits of the opposite algebra category are computed by algebra pushouts. $\square$

All the fibers in this lemma are homotopy fibers. For the trivial group acting on X the resulting object is $\Omega_x X$; ordinary set-theoretic stabilizers would give a different construction.

Proof of Theorem 7.1. The filtered stabilizer. The based loop object with function algebra $D = \text{Bar}_k(A)$ acts on the fiber with function algebra $C = B \otimes_A k$. In algebra terms this gives a right coaction $\delta : C \rightarrow C \otimes D$ with

$$f = (\varepsilon_C \otimes \text{id})\delta, \quad \Delta_D f = (f \otimes \text{id})\delta. \quad (7.4)$$

These identities belong to the action diagram with its coherent structure maps.

Place D in constant filtration. The coaction lifts to

$$\tilde{C} \longrightarrow \tilde{C} \otimes_{\text{Day}} \text{step}_0(D) :$$

its zero layer is the unit $k \rightarrow D$, and its positive layers are δ . Unit compatibility supplies the compatibility between the layers. The point and orbit map lift as well. Lemma 7.2 therefore constructs the filtered Hopf algebra

$$\tilde{E} = \text{step}_0(D) \otimes_{\tilde{C}} \text{step}_0(k). \quad (7.5)$$

Since the monoidal structure is Day convolution, its coproduct has the required sum filtration.

The canonical bar coproduct. Forgetting filtration preserves this construction. The stabilizer of the chosen point of the fiber of $\text{Spec } B \rightarrow \text{Spec } A$ under $\Omega \text{Spec } A$ is $\Omega \text{Spec } B$. More explicitly, restricting the fiber-action groupoid at its basepoint identifies its arrow object and composition with $* \times_{\text{Spec } B} *$ and its loop composition. Hence the forgotten Hopf structure is the canonical structure on $k \otimes_B k$, with the asserted map from D .

The ordinary bar skeletons. To identify the filtration, compute (7.5) by the two-sided bar in filtered modules. Its j -th term is

$$\text{step}_0(D) \otimes_{\text{Day}} \tilde{C}^{\otimes_{\text{Day}} j} \otimes_{\text{Day}} \text{step}_0(k).$$

Expanding (7.3), a term with r ideal factors in specified positions is $\text{step}_r(D \otimes I^{\otimes r})$. Degeneracy maps insert units. Thus the nondegenerate j -th layer is exactly $\text{step}_j(D \otimes I^{\otimes j})$. At filtration s , all summands in simplicial degrees $j \leq s$ are present, so these degrees agree with $B_j(D, C, k)$. For $j > s$, each remaining summand has a unit in some position; it is generated by degeneracies from lower degrees. The end faces remain f and ε_C , and the inner faces remain the multiplication of C . The realization at level s is consequently the ordinary s -skeleton with its original attaching maps: both the retained terms and the maps attaching them have been identified. Its successive quotient is $\Sigma^s(D \otimes I^{\otimes s})$. This identifies every layer of $\tilde{E}$ and finishes the proof. $\square$

7.2 The zero-column coaction

From now on in this section $k = H\mathbb{F}_2$, and tensor products of homotopy groups are over $\mathbb{F}_2$. Set

$$R = \pi_* C, \quad H = \pi_* D, \quad K = \text{im}(R \xrightarrow{f_*} H), \quad Q = H/HK^+,$$

where $K^+ = \ker(K \rightarrow \mathbb{F}_2)$. Write $p : H \twoheadrightarrow Q$ for this quotient and $\bar{R} = \ker(R \rightarrow \mathbb{F}_2)$. Equation (7.4) makes $\Delta K \subset K \otimes H$. The quotient Q is the zero column after taking d_1 -homology. The next proposition computes its coaction on the other columns. The specific formula matters: it will let us realize the coinvariants by a subcomplex whose power operations can be checked directly.

Proposition 7.3. *The coaction of Q obtained by projecting the first factor of the actual E^2 -page coproduct to its zero column is induced by the following chain coaction on the algebraic bar:*

$$\begin{aligned} H \otimes \bar{R}^{\otimes s} &\longrightarrow Q \otimes H \otimes \bar{R}^{\otimes s}, \\ h[c_1 | \cdots | c_s] &\longmapsto \sum p(h_{(1)}) \otimes h_{(2)}[c_1 | \cdots | c_s]. \end{aligned} \quad (7.6)$$

Thus the actual coaction uses the coproduct of the H coefficient and the trivial coaction on the bar letters.

Proof. The double bar. We compare the stabilizer coproduct with a filtered double-bar model whose terms are

$$X_{i,j} = D \otimes C^{\otimes i} \otimes C \otimes C^{\otimes j}.$$

The central C carries the unit filtration, in addition to the two ordinary skeletal filtrations. The horizontal augmentation gives a filtered equivalence to $B_{\bullet}(D, C, k)$. To see this at the level of filtered objects, form both bar constructions using $\tilde{C}$. The horizontal bar

$$B_{\bullet}(\text{step}_0(D), \tilde{C}, \tilde{C}) \longrightarrow \text{step}_0(D)$$

has the usual extra degeneracy: move the central factor into the last horizontal bar slot and put the unit in its place. An ideal factor has filtration one before and after this move, so the contraction preserves filtration. Vertical realization gives the asserted equivalence.

There is a bisimplicial map to $B_i(D, C, k) \otimes B_j(D, C, k)$ formed from Δ_D, δ, f . In element notation for these actual compositions of algebra maps, it is

$$\begin{aligned} & h[c_1 | \cdots | c_i] a[c_{i+1} | \cdots | c_{i+j}] \\ & \longmapsto \sum h_{(1)}[c_{1(0)} | \cdots | c_{i(0)}] \otimes h_{(2)} \left(\prod_{\ell=1}^i c_{\ell(1)} \right) f(a)[c_{i+1} | \cdots | c_{i+j}]. \end{aligned} \quad (7.7)$$

The horizontal first face is compatible by the second identity in (7.4). The horizontal last face uses its first identity, and the vertical first face uses multiplicativity of f . Inner faces and degeneracies use multiplicativity and unitality of the coaction. Functoriality of the two bar constructions supplies the coherent simplicial map. The central ideal factor can lower filtration under f , but cannot raise it, so this map is filtered.

Agreement with the stabilizer coproduct. First realize (7.7) horizontally, using throughout the filtered algebras $\tilde{C}, \tilde{E}, \text{step}_0(D)$. The resulting map on the horizontal augmentation is

$$\lambda_D = (q \otimes \text{id}) \Delta_D : \text{step}_0(D) \longrightarrow \tilde{E} \otimes_{\text{Day}} \text{step}_0(D).$$

Its $\tilde{C}$ -linearity uses the specified pushout homotopy $qf \simeq \eta \varepsilon_C$, together with (7.4); explicitly,

$$(q \otimes \text{id}) \Delta_D f = (qf \otimes \text{id}) \delta \simeq 1 \otimes f.$$

In the opposite category, this is the left action of the homotopy stabilizer on G : the path $xh \simeq x$ identifies the orbit points of hg and g . Realizing vertically, or base changing along the chosen point, restricts this action to the stabilizer itself. It is its regular multiplication, hence on functions its coproduct. This verifies the structure being computed, including the point homotopy that controls the comparison.

The zero-column component. To compute this component, we describe the inverse of the double-bar equivalence on associated gradeds. In weight s , the central-unit cuts give $s+1$ copies of $\Sigma^s(D \otimes I^{\otimes s})$, while central-ideal cuts give s copies in degree shifted down by one. The same-weight attaching matrix is

$$(\alpha_0, \dots, \alpha_s) \longmapsto (\alpha_0 + \alpha_1, \dots, \alpha_{s-1} + \alpha_s),$$

with each entry acting on the corresponding copy of $D \otimes I^{\otimes s}$. It is the incidence matrix of an interval: its kernel is the diagonal copy and its cokernel is zero. Projection to an end cut identifies the kernel with one copy. At $s=0$ there is only that copy. Over a general base the same calculation uses the usual orientation signs.

The inverse of the end-cut projection on weight s is therefore the sum of all unit cuts. Applying (7.7) gives the twisted deconcatenation formula. Its component with first bar

degree zero retains only the cut $i = 0$. A positive prefix that becomes degenerate has lower total filtration and contributes nothing to this weight. The zero-column component on the associated graded thus acts on h by Δ_H and leaves the letters alone.

Passage to E^2 . It remains to check compatibility with d_1 . Projection to the E^1 zero column H is not generally a chain map, because d_1 enters that column. For $r \in R$, the coideal condition and the orbit identity give

$$(p \otimes \text{id})\Delta_H f_*(r) = 1 \otimes f_*(r).$$

After this quotient, all faces commute with the displayed coaction; the incoming zero-column boundaries are also killed. The associated-graded calculation consequently induces exactly (7.6) on E^2 . $\square$

7.3 Finite-page Hopf structures

Index the spectral sequence by

$$E_{s,t}^1 = (H \otimes \overline{R}^{\otimes s})_t, \quad d_r : E_{s,t}^r \longrightarrow E_{s-r,t+r-1}^r.$$

The bar index s is nonnegative; the internal index t is unrestricted. Over $\mathbb{F}_2$, the spectral sequence of the Day tensor product is the tensor product spectral sequence. This follows on E^1 from the displayed associated gradeds and the field Künneth formula, and on later pages from the Künneth formula for complexes. For fixed total bar degree the decomposition into two bar degrees is finite. Internal degrees use ordinary direct-sum tensor products; no completion is involved.

Theorem 7.1 therefore makes each finite page a graded commutative Hopf algebra, with d_r a derivation and a coderivation. Its zero column is a Hopf subalgebra, and projection onto that column is a Hopf retraction *on the page*. It is generally not a map of d_r -complexes. The next section uses descent to handle this failure of chain compatibility.

8 Primitive classes and Hopf-algebra descent

For an augmented $\mathbb{E}_\infty$ -ring $A \rightarrow \mathbb{S}$, write

$$\text{Bar}(A) = \text{Bar}_{\mathbb{S}}(A) = \mathbb{S} \otimes_A \mathbb{S}.$$

Its loop-group coproduct makes $H_*(\text{Bar}(A); \mathbb{F}_2)$ a commutative $\mathbb{Z}$ -graded Hopf algebra. A homogeneous element z is *primitive* if $\Delta z = z \otimes 1 + 1 \otimes z$.

Theorem 8.1. *Let $A \rightarrow B$ be an Adams-type map of augmented $\mathbb{E}_\infty$ -rings over $\mathbb{S}$. Then*

$$H_*(\text{Bar}(A); \mathbb{F}_2) \longrightarrow H_*(\text{Bar}(B); \mathbb{F}_2)$$

reflects nilpotence on homogeneous primitive elements of nonzero degree. No boundedness assumption is imposed on A or B .

The proof reduces the claim to the zero column of each finite page of the relative bar spectral sequence. By Lemma 8.9, a class from filtration zero that vanishes in the target already vanishes on a finite page. It therefore suffices to keep every power of a nonnilpotent primitive nonzero in these zero columns.

The sphere-flat algebra $B \otimes_A \mathbb{S}$ of Lemma 6.1 has vanishing nonzero-index operations by Proposition 6.2. Using the filtered Hopf structure of Theorem 7.1, Lemma 8.8 transfers this vanishing to the canonical Hopf-module coinvariants on the second page. Proposition 8.6 preserves it on later pages by realizing the new coinvariants as the homology of an actual subcomplex. Coideal descent and the square operation then show that each zero-column quotient reflects primitive nilpotence.

8.1 Coideals and relative Hopf modules

Throughout this subsection, tensor products are over a field k unless another base is indicated. A coaugmented coalgebra is *conilpotent* if each element of its coaugmentation coideal is annihilated by a sufficiently long iterated reduced coproduct. In particular, its coradical is the one-dimensional coaugmentation. Our convention for a right coideal subalgebra $K \subset Q$ is

$$\Delta(K) \subset K \otimes Q.$$

We write $K^+ = \ker(\varepsilon : K \rightarrow k)$, and use S and η for the antipode and unit of a Hopf algebra. No assertion that K is a Hopf subalgebra is implicit in this notation.

Lemma 8.2. *Let Q be a conilpotent Hopf algebra over k , and let $K \subset Q$ be a right coideal subalgebra. Then Q is free and faithfully flat as a right K -module.*

Proof. We give the ordinary Hopf-algebra argument underlying [AAGMV, Lemma 3.5 and Proposition 3.6(a)]. This also makes clear that no algebraic-closure or characteristic-zero assumption on k is needed here.

A relative Hopf module is a right K -module M with a right Q -coaction satisfying

$$\rho(mk) = \sum m_{(0)}k_{(1)} \otimes m_{(1)}k_{(2)}.$$

Every nonzero Q -comodule has a nonzero socle, and, since Q is connected, that socle consists of coinvariants. Indeed, a nonzero comodule contains a nonzero finite-dimensional subcomodule; a simple subcomodule of it must be the trivial comodule because the coradical of Q is k . Thus, for $M \neq 0$,

$$X = M^{\text{co}Q} = \{m \in M : \rho(m) = m \otimes 1\}$$

is nonzero.

Give $X \otimes K$ the coaction that is trivial on X and is the restriction of Δ on K . Its socle is $X \otimes k$: the coinvariants in K are k , as follows by applying the counit to $\Delta(k) = k \otimes 1$. The action map

$$X \otimes K \longrightarrow M, \quad x \otimes k \longmapsto xk$$

is a morphism of relative Hopf modules and restricts to the identity on X . Its kernel is zero. Otherwise that kernel would be a nonzero Q -comodule with nonzero socle, contradicting injectivity on $X \otimes k$. Consequently every nonzero relative Hopf module contains a nonzero relative Hopf submodule that is free over K .

To exhaust M , consider pairs (N, X) in which $N \subset M$ is a relative Hopf submodule, $X \subset N$ is a vector subspace, and multiplication is an isomorphism $X \otimes K \rightarrow N$. Order these pairs by inclusion in both entries. Unions of chains are again such pairs, so a maximal pair exists. If $N \neq M$, apply the preceding paragraph to M/N . It supplies a nonzero subspace $Y \subset M/N$ such that $Y \otimes K$ maps isomorphically onto a relative Hopf submodule. Choose a vector-space lift $\tilde{Y} \subset M$ of Y and put

$$\tilde{X} = X \oplus \tilde{Y}, \quad \tilde{N} = N + \tilde{Y}K.$$

The module $\tilde{N}$ is precisely the inverse image of YK in M , hence is also a subcomodule. Multiplication $\tilde{X} \otimes K \rightarrow \tilde{N}$ is surjective. It is injective because a relation first projects to a relation in the free module $Y \otimes K$ and then to one in $X \otimes K$. This strictly enlarges the maximal pair, a contradiction. Thus M is free over K . Taking $M = Q$ with its regular coaction proves freeness. The free module has nonzero rank, so it is faithfully flat. $\square$

The next statement uses commutativity. In our application the field is $\mathbb{F}_2$, so forgetting the internal grading of a graded commutative Hopf algebra introduces no signs.

Lemma 8.3. *Let Q be a commutative conilpotent Hopf algebra over k and $K \subset Q$ a right coideal subalgebra. Put*

$$Q' = Q/QK^+, \quad q : Q \longrightarrow Q'.$$

Then Q' is a conilpotent Hopf algebra and

$$\begin{aligned} \beta : Q \otimes_K Q &\xrightarrow{\sim} Q' \otimes Q, \\ a \otimes b &\longmapsto \sum q(a_{(1)}) \otimes a_{(2)}b \end{aligned} \tag{8.1}$$

is an isomorphism. Moreover,

$$Q^{\text{co}Q'} = K, \quad \lambda_Q = (q \otimes \text{id})\Delta. \tag{8.2}$$

Let $\mathcal{D}(Q, Q')$ be the category of left Q -modules M with a left Q' -coaction $\lambda_M(m) = \sum m_{(-1)} \otimes m_{(0)}$ satisfying

$$\lambda_M(am) = \sum q(a_{(1)})m_{(-1)} \otimes a_{(2)}m_{(0)}. \tag{8.3}$$

There is an equivalence

$$\text{Mod}_K \simeq \mathcal{D}(Q, Q'), \quad N \longmapsto Q \otimes_K N, \tag{8.4}$$

whose inverse is $M \mapsto M^{\text{co}Q'}$. In particular, coinvariants are exact on $\mathcal{D}(Q, Q')$.

Proof. For $k \in K$, right coideality gives

$$(q \otimes \text{id})\Delta(k) = \sum \varepsilon(k_{(1)}) \otimes k_{(2)} = 1 \otimes k.$$

It follows that QK^+ is a biideal. The quotient coalgebra remains conilpotent. A connected bialgebra has an antipode, obtained by the locally finite convolution series in $\text{id} - \eta\varepsilon$; hence Q' is a Hopf algebra and q is a Hopf map.

The displayed identity on K makes (8.1) balanced over K . An inverse is

$$q(a) \otimes b \longmapsto \sum a_{(1)} \otimes_K S(a_{(2)})b. \tag{8.5}$$

To check that this descends through q , replace a by ak with $k \in K^+$. In the resulting expression, $k_{(1)}$ belongs to K , so it can be moved across the balanced tensor product. The remaining factor is $\sum k_{(1)}S(k_{(2)}) = \varepsilon(k) = 0$. The two composites of (8.1) and (8.5) are identities by the antipode identities. This verification uses a right coideal subalgebra, not an antipode on K .

By Lemma 8.2, Q is faithfully flat over K . The equalizer of $Q \rightrightarrows Q \otimes_K Q$ is therefore K . Under β , the two maps become λ_Q and $a \mapsto 1 \otimes a$, proving (8.2).

Faithfully flat module descent [Stacks, Proposition 35.3.9, Tag 023N] applies to the commutative ring map $K \rightarrow Q$. Under (8.1), its descent datum on a Q -module is exactly a coaction satisfying (8.3): the coaction counit gives normalization, and coassociativity gives the descent cocycle condition. The descended module is the equalizer defining $M^{\text{co}Q'}$, and scalar extension gives its inverse. This proves (8.4). As an equivalence of abelian categories, it has an exact inverse. Exactness here concerns Q' -comodules with the compatible Q -module structure (8.3); it is not a statement about arbitrary Q' -comodules. $\square$

Corollary 8.4. *Suppose $k = \mathbb{F}_2$, and let Q, K, Q' be as in Lemma 8.3, with compatible $\mathbb{Z}$ -gradings. Assume Q has operations Q^i satisfying $Q^{|a|}a = a^2$ on homogeneous elements, and that $Q^iK = 0$ for every nonzero integer i . Then $Q \rightarrow Q'$ reflects nilpotence on homogeneous primitives of nonzero degree.*

Proof. If a primitive z has nilpotent image, choose a power of two $w = z^{2^n}$ with $q(w) = 0$. It is still primitive, so

$$(q \otimes \text{id})\Delta(w) = 1 \otimes w.$$

Equation (8.2) puts w in K . Its degree $2^n|z|$ is nonzero, and hence $w^2 = Q^{|w|}w = 0$. Thus z is nilpotent. $\square$

8.2 Vertical operations on the relative bar spectral sequence

We now work over $k = \mathbb{F}_2$, and use Hk for its Eilenberg–Mac Lane spectrum. If $T_\bullet$ is a simplicial $\mathbb{E}_\infty Hk$ -algebra, its skeletal filtration has a homological spectral sequence. We use the normalized first page and the indexing

$$P_1^{s,t} = N_s \pi_t T_\bullet, \quad d_r : P_r^{s,t} \longrightarrow P_r^{s-r, t+r-1}. \quad (8.6)$$

The total degree is $s + t$, and $s \geq 0$. Internal degrees t may be unbounded in both directions.

Lemma 8.5. *On every finite page of (8.6) there are additive vertical operations*

$$Q^i : P_r^{s,t} \longrightarrow P_r^{s, t+i} \quad (i \in \mathbb{Z}).$$

They are the levelwise Dyer–Lashof operations on the first page, commute with d_r , and induce the operations on P_{r+1} . They satisfy the Cartan formula for the shuffle product. On the zero column they satisfy $Q^{|a|}a = a^2$. These assertions about finite pages do not require convergence of the full spectral sequence.

Proof. For a proper simplicial model, the operations and differential identity are those of [BM, Theorem 6.1(i)–(ii)]. Proper models can be chosen as explained there. We specify why the finite-page assertions apply with the present grading and without a lower bound, rather than invoking the convergence assertion accompanying that theorem.

For a degree- t class, Q^i uses the cell of index $i - t$ in its extended square; it is zero when $i < t$. Thus a negative value of i is permitted when $i \geq t$. The construction uses spheres $\Sigma^t Hk$, which exist for every integer t . For any fixed operation, an $\mathbb{E}_n$ structure with n sufficiently large lies in the range of the cited theorem; the given $\mathbb{E}_\infty$ structure provides all these structures.

More explicitly, the proof of the differential identity in [BM, proof of Theorem 6.1, pp. 18–20] represents a class surviving to $P_r^{s,t}$ by a simplicial Hk -module whose nondegenerate terms consist of the sphere $\Sigma^t Hk$ in degree s , a finite succession of cones in degrees $s - 1, \dots, \max(0, s - r + 1)$, and, when $r \leq s$, the sphere $\Sigma^{t+r-1} Hk$ in degree $s - r$. The cone range is empty when its lower endpoint exceeds $s - 1$, in particular when $s = 0$. For $r \leq s$, the last sphere represents the differential; applying the extended-power construction to this finite model proves $Q^i d_r = d_r Q^i$.

When $r > s$, there is no outgoing differential. Incoming boundaries still come from column $s + r$, where the preceding differential identity applies and shows that the operations preserve boundaries. The construction uses only these spheres and cones and their finite nullhomotopies. Shifting their internal degrees to negative integers changes none of the identities. Thus the operations descend through d_1 and, inductively, through each finite-page differential. No assertion about the abutment is needed.

For completeness, Cartan for the product in (8.6) follows from a different, elementary calculation. If $a \in P_1^{s,t}$ and $b \in P_1^{u,v}$, their product is

$$a \star b = \sum_{\sigma} \mu(s_{\sigma} a, t_{\sigma} b),$$

where the sum is over (s, u) -shuffles and the two operators in each summand are composites of degeneracies. Degeneracies are algebra maps. Levelwise additivity, naturality, and Cartan give

$$\begin{aligned} Q^i(a \star b) &= \sum_{\sigma} \sum_{j+\ell=i} \mu(s_{\sigma} Q^j a, t_{\sigma} Q^{\ell} b) \\ &= \sum_{j+\ell=i} Q^j a \star Q^{\ell} b. \end{aligned}$$

The sums are finite: instability gives $j \geq t$ and $\ell \geq v$, so $t \leq j \leq i - v$. The operations preserve degeneracies and the normalized subspaces, so the calculation applies to the normalized first page. Both multiplication and the operations on the next page are induced on homology. Cartan therefore persists on every finite page.

The usual square identity holds on $\pi_* T_0$. The zero column has no outgoing differential, and each later zero column is a quotient of the preceding one. Both the product and the operations descend, proving the square identity there. $\square$

The vertical operations preserve bar degree; the shuffle product adds bar degrees. Thus, in positive bar degree, the square identity for a levelwise operation must not be identified with a shuffle square. We use the square identity only in the zero column. Nor do we need the vertical operations to commute with the filtered Hopf coproduct.

8.3 Descent from one Hopf page to the next

We separate each Hopf page into its zero column and the canonical coinvariants for projection onto that column. Let P be a commutative bigraded Hopf algebra over $\mathbb{F}_2$, with nonnegative first degree, and let $Q = P^{0,*}$ be its zero column. Projection $\pi : P \rightarrow Q$ and inclusion $Q \rightarrow P$ are Hopf maps: the coproduct preserves the sum of the first degrees. Set

$$\rho = (\pi \otimes \text{id})\Delta, \quad V = P^{\text{co}Q}.$$

Thus V is the subalgebra of coinvariants for the specified coaction ρ . Multiplication induces an isomorphism

$$Q \otimes V \xrightarrow{\sim} P. \quad (8.7)$$

One can see this directly by defining

$$\vartheta(p) = \sum S(\pi(p_{(1)}))p_{(2)}.$$

The antipode identities and coassociativity give $\rho(\vartheta(p)) = 1 \otimes \vartheta(p)$. The inverse of multiplication is

$$p \longmapsto \sum \pi(p_{(1)}) \otimes \vartheta(p_{(2)}).$$

Composing with multiplication gives p by the antipode identity; on av , with $a \in Q$ and $v \in V$, the reverse composite is $a \otimes v$. This proves (8.7). It is an algebra isomorphism because P is commutative and V is a subalgebra. Also $V^{0,*} = k$: coinvariants of the regular coaction on Q are its scalars.

Proposition 8.6. *Suppose $(P_r, d_r)_{r \geq 2}$ is a spectral sequence of commutative bigraded Hopf algebras over $\mathbb{F}_2$, with nonnegative bar degree, differential as in (8.6), and coproduct preserving total bar degree. Assume that d_r is a derivation and a coderivation, and that the augmentation commutes with it. Suppose the vertical operations have the properties in Lemma 8.5. Write*

$$Q_r = P_r^{0,*}, \quad V_r = P_r^{\text{co}Q_r}, \quad P_r = Q_r \otimes V_r$$

using (8.7). If Q_2 is conilpotent and

$$Q^i V_2 = 0 \quad (i \neq 0), \quad Q^0 V_2 \subset V_2, \quad (8.8)$$

then Q_r is conilpotent and (8.8) holds with V_r in place of V_2 for every finite $r \geq 2$. Every map $Q_r \rightarrow Q_{r+1}$ reflects nilpotence on homogeneous primitive elements of nonzero degree.

Proof. Fix a page r on which the assertions hold, and abbreviate $P = P_r$, $Q = Q_r$, $V = V_r$, and $d = d_r$. Let $K \subset Q$ be the subalgebra generated by the homogeneous elements

$$d(v), \quad v \in V^{r,*}. \quad (8.9)$$

This is the transgression algebra for the page: its generators are the differentials from V into the zero column. All these elements have augmentation zero. Since d vanishes on Q and is Q -linear, (8.7) identifies the incoming boundaries in the zero column as QK^+ . Hence

$$Q_{r+1} = Q' = Q/QK^+. \quad (8.10)$$

To apply Corollary 8.4 to this quotient, we will show that K is a right coideal and has vanishing nonzero-index operations. To continue the induction, we will then realize the canonical V_{r+1} as the homology of the subcomplex $K \otimes V \subset P$. This inclusion will carry the operation vanishing to the next page.

The transgression coideal. Coassociativity implies

$$\Delta(V) \subset V \otimes P. \quad (8.11)$$

Indeed, applying ρ to the first factor of $\Delta(v)$, for $v \in V$, gives $1 \otimes \Delta(v)$; tensoring over a field preserves the equalizer defining V . If v has bar degree r , the component of $\Delta(v)$ with first bar degree zero is $1 \otimes v$, since $V^{0,*} = k$. Its component with second bar degree zero is a finite sum $\sum_j v_j \otimes q_j$, with $v_j \in V^{r,*}$ and $q_j \in Q$. The remaining components have both bar degrees strictly between 0 and r , so their differentials vanish. The coderivation identity therefore gives

$$\Delta(dv) = 1 \otimes dv + \sum_j d(v_j) \otimes q_j. \quad (8.12)$$

Thus K is a right coideal subalgebra. Lemma 8.3 shows that Q' is a conilpotent Hopf algebra.

Operation commutation and the induction hypothesis give

$$Q^i(dv) = d(Q^i v) = 0 \quad (i \neq 0), \quad Q^0(dv) = d(Q^0 v) \in K.$$

The latter inclusion uses both the preservation of bar degree and the Q^0 -stability of V . Cartan extends these assertions from the generators (8.9) to K .

The subcomplex $K \otimes V$. The projection $P \rightarrow Q$ need not commute with d , so one cannot identify V_{r+1} with $H(V, d)$. Instead put $q : Q \rightarrow Q'$ and use

$$\rho' = (q\pi \otimes \text{id})\Delta : P \longrightarrow Q' \otimes P. \quad (8.13)$$

It is a chain coaction. The coderivation identity gives

$$\rho' d - (\text{id} \otimes d)\rho' = (q\pi d \otimes \text{id})\Delta = 0.$$

For the last equality, πd is supported on inputs of bar degree r , and its image is exactly the boundary ideal killed in (8.10). Moreover P remains a Q -module, d is Q -linear, and (8.13) satisfies (8.3). To apply Lemma 8.3, forget the internal grading and separate the columns by their residue modulo r . For each $0 \leq j < r$, the terms $\bigoplus_t P^{j+nr,t}$, $n \geq 0$, are ordinary Q -modules because Q has bar degree zero. The differential lowers n by one, and the coaction preserves each term. They therefore form a complex in $\mathcal{D}(Q, Q')$, with no bound on t required.

Under the splitting $P = Q \otimes V$, this coaction is the quotient regular coaction on Q and the trivial coaction on V . Consequently its coinvariant subcomplex is the actual subspace

$$P^{\text{co}Q'} = K \otimes V \subset P, \quad (8.14)$$

where the inclusion is multiplication. In particular d preserves this subspace. Exactness in the relative Hopf-module category, applied to cycles and boundaries, gives

$$H(K \otimes V, d) \xrightarrow{\sim} H(P, d)^{\text{co}Q'}. \quad (8.15)$$

The map is induced by inclusion of cycles. It remains to identify its target with the canonical coinvariants on the next page.

The canonical coinvariants on the next page. For a generator $a = d(v)$ of K^+ , multiplication by v is a chain homotopy for multiplication by a :

$$d(vp) + v d(p) = ap.$$

Thus QK^+ acts trivially on $H(P, d)$, and the Q -action there factors through Q' . The chain map $q\pi : P \rightarrow Q'$ induces the zero-column projection on the next page: it vanishes in positive bar degree and is the quotient map in bar degree zero. Taking homology in (8.13) therefore gives exactly the coaction defined by the next-page Hopf projection. The right side of (8.15) is consequently V_{r+1} .

Both K and V have vanishing Q^i for $i \neq 0$ and are preserved by Q^0 . Cartan gives the same properties on the subcomplex (8.14). Since (8.15) is induced by cycle inclusion, they pass to the canonical subspace V_{r+1} . This proves the induction. It uses no compatibility between the vertical operations and Δ .

Finally, Corollary 8.4 applies to $Q \rightarrow Q'$, using the operation vanishing on K and the square identity in the zero column. It proves the asserted nilpotence reflection on this page and hence on every page. Every map used above has its specified bidegree, so the descent identifications restrict to the original homogeneous components. $\square$

8.4 The initial coinvariants of the relative bar construction

We verify that the input in Proposition 8.6 is the canonical input for the map in Theorem 8.1.

Lemma 8.7. *For an augmented $\mathbb{E}_\infty$ -H $\mathbb{F}_2$ -algebra T , the Hopf algebra $\pi_*(H\mathbb{F}_2 \otimes_T H\mathbb{F}_2)$ is conilpotent.*

Proof. Apply the bar filtration of Theorem 7.1 to $H\mathbb{F}_2 \rightarrow T$. Its zeroth stage is $H\mathbb{F}_2$, and its coproduct preserves the sum filtration. After projection to the augmentation ideal, the zeroth stage is zero. Thus the reduced m -fold coproduct is zero on filtration stage N whenever $m > N$: each surviving tensor factor would have positive filtration, whereas their sum is at most N . Every homotopy class comes from some finite stage, because the bar construction is the filtered colimit of its skeletons and spheres in $H\mathbb{F}_2$ -modules are compact. Each reduced class is therefore annihilated by a sufficiently long reduced coproduct. Internal degrees play no role in this argument. $\square$

For the remainder of the section, let $A \rightarrow B$ satisfy the hypotheses of Theorem 8.1 and put

$$C = (B \otimes_A \mathbb{S}) \otimes_{\mathbb{S}} H\mathbb{F}_2, \quad D = \text{Bar}(A) \otimes_{\mathbb{S}} H\mathbb{F}_2, \quad E = \text{Bar}(B) \otimes_{\mathbb{S}} H\mathbb{F}_2.$$

The relative bar presentation and its specified maps are those of Theorem 7.1:

$$E \simeq D \otimes_C H\mathbb{F}_2.$$

Write

$$R = \pi_* C, \quad H = \pi_* D, \quad K = \text{im}(R \rightarrow H), \quad Q = H/HK^+. \quad (8.16)$$

The augmentation ideal of R will be denoted $\bar{R}$.

Lemma 8.8. *The zero column on the second page is Q , and the canonical Hopf-module splitting is*

$$P_2 \cong Q \otimes V_2, \quad V_2 = \text{Tor}_*^R(K, \mathbb{F}_2) = P_2^{\text{co} Q}. \quad (8.17)$$

The subspace V_2 is preserved by Q^0 , and $Q^i V_2 = 0$ for every nonzero integer i . The map $H \rightarrow Q$ reflects nilpotence on homogeneous primitives of nonzero degree.

Proof. We first apply coideal descent to $H \rightarrow Q$ to obtain primitive nilpotence reflection. We then identify the Tor factor in (8.17) with the canonical coinvariants by using the specified coaction on the bar complex. Realizing that factor by an inclusion of cycles will also give its operation properties.

The zero-column quotient. The actual D -coaction on C and the orbit map $C \rightarrow D$ make K a right coideal subalgebra of H . Lemma 8.7 makes H conilpotent. By Lemma 6.1 and Proposition 6.2, all nonzero-index operations on R vanish. Naturality gives the same vanishing on K , and Q^0 preserves K . Lemmas 8.2 and 8.3 apply to $K \subset H$: H is faithfully flat over K , and Q is a conilpotent Hopf algebra. Corollary 8.4 already proves the last assertion of the lemma.

The Tor factor. The normalized first page of the relative bar spectral sequence is the algebraic bar complex

$$P_1^{s,*} = H \otimes \bar{R}^{\otimes s}. \quad (8.18)$$

Because H is flat over K , scalar extension from the corresponding bar complex with coefficient K gives

$$\text{Tor}_*^R(H, \mathbb{F}_2) \cong H \otimes_K \text{Tor}_*^R(K, \mathbb{F}_2). \quad (8.19)$$

The K -action on $\text{Tor}_*^R(K, \mathbb{F}_2)$ factors through its augmentation. To see this, lift an element of K to R . The two R -actions on the derived tensor product $K \otimes_R^{\mathbb{L}} \mathbb{F}_2$ agree; the action from the second factor is the augmentation action on $\mathbb{F}_2$. Surjectivity of $R \rightarrow K$ proves the assertion. Thus the right side of (8.19) is $Q \otimes V_2$ with $V_2 = \text{Tor}_*^R(K, \mathbb{F}_2)$.

The canonical coinvariants. To identify this Tor factor with the specified coinvariants, put $q : H \rightarrow Q$. The zero-cut comparison in Proposition 7.3 identifies the coaction defining the second-page Hopf-module splitting with that induced on (8.18) by

$$h[c_1 | \cdots | c_s] \mapsto \sum q(h_{(1)}) \otimes h_{(2)}[c_1 | \cdots | c_s]. \quad (8.20)$$

Here the $\bar{R}$ factors have trivial coaction. For $k \in K$, right coidealness gives $(q \otimes \text{id})\Delta(k) = 1 \otimes k$. Consequently

$$K \otimes \bar{R}^{\otimes s} \subset H \otimes \bar{R}^{\otimes s} \quad (8.21)$$

is a subcomplex of coinvariants. The comparison (8.19) is colinear when its coefficient K complex is given this trivial coaction. After passage to homology it becomes the regular coaction on Q and the trivial coaction on V_2 . Its coinvariants are exactly $1 \otimes V_2$, proving (8.17).

In particular, the canonical V_2 is realized by inclusion of cycles from (8.21). This inclusion is injective on homology: under (8.19) it is the map $V_2 \rightarrow H \otimes_K V_2$, which is injective by faithful flatness, and then identifies with $v \mapsto 1 \otimes v$ in $Q \otimes V_2$. Both R and K have vanishing nonzero-index operations and are preserved by Q^0 . The augmentation ideal $\bar{R}$ is also operation-stable by naturality of the augmentation. Levelwise Cartan therefore gives the same vanishing and stability on (8.21). Lemma 8.5 passes these properties to its homology and to its image, which is precisely the canonical V_2 . $\square$

8.5 Finite skeletons and the single-step theorem

Lemma 8.9. *Let $F_0X \rightarrow F_1X \rightarrow \dots$ be an exhaustive increasing filtration of an $H\mathbb{F}_2$ -module, with no assumption on the internal degrees of its filtration quotients. If $a \in \pi_*F_0X$ maps to zero in π_*X , then its zero-column image is zero on some finite page of the spectral sequence of $F_\bullet X$.*

Proof. Compactness of suspensions of $H\mathbb{F}_2$ gives $\pi_*X = \text{colim}_N \pi_*F_NX$. Thus a is already zero in π_*F_NX for some N . Give F_NX the finite filtration $F_{\min(s,N)}X$. Its spectral sequence has finitely many columns and computes the filtration on π_*F_NX without a boundedness hypothesis on the individual quotients. Its limiting zero column is the image of π_*F_0X in π_*F_NX , so the image of a there is zero. The zero column has no outgoing differentials. Consequently the zero-column class of a becomes zero after finitely many incoming differentials, unless it was already zero on the first page. The inclusion of this truncated filtration in $F_\bullet X$ induces a map of spectral sequences. Naturality sends that zero class to the zero-column image of a on the same finite page of the full spectral sequence. $\square$

Proof of Theorem 8.1. Use the relative bar filtration of $E = D \otimes_C H\mathbb{F}_2$ with the notation (8.16). By Theorem 7.1, it is a filtered Hopf algebra with the sum filtration on tensor products, and it induces the canonical Hopf map $D \rightarrow E$. Over the field $\mathbb{F}_2$, the associated spectral sequence of a tensor product filtration is the tensor product spectral sequence: this follows first from Künneth on the associated graded, and then page by page from Künneth for complexes of vector spaces. Each fixed bar degree has only finitely many decompositions into two nonnegative bar degrees. It follows that every page is a Hopf algebra and that its differential is both a derivation and a coderivation. Lemma 8.5 supplies the required vertical operations.

Lemma 8.8 identifies the initial canonical coinvariants and verifies their operation properties, as well as conilpotence of the zero column. Proposition 8.6 therefore applies to all finite pages. If $z \in H = \pi_*D$ is a nonnilpotent homogeneous primitive of nonzero degree, its image in Q_2 is nonnilpotent by Lemma 8.8, and its image in every subsequent Q_r remains nonnilpotent by the proposition. All these images are primitive because the zero-column quotient maps are Hopf maps.

Suppose the image of z in π_*E were nilpotent. Choose a power of two z^{2^n} whose image is zero. This is a class in the zeroth filtration stage $D = F_0E$, and it has nonzero image in the zero column on every finite page. Lemma 8.9 gives a contradiction. The powers here are powers of a class already in F_0E , so no multiplicative extension in positive filtration is involved. The argument uses finite skeletons and does not identify survival on all pages with survival in an unbounded abutment. Thus the image of z in π_*E is nonnilpotent, as required. $\square$

9 The commutative transfinite obstruction

Theorem 9.1. *The augmentation*

$$\mathrm{Sym}_{\mathbb{S}}(\mathbb{S}) \longrightarrow \mathbb{S}$$

which sends the degree-zero generator to zero is not a continuous transfinite composite of Adams-type maps in commutative ring spectra.

Proof. We construct a nonnilpotent primitive of degree one in the source bar algebra. Theorem 8.1 preserves its nonnilpotence at each Adams step, and filtered colimits preserve it at limit stages. The contradiction will come from the absence of degree-one homology in the bar of the target sphere.

The initial primitive. The free commutative algebra functor preserves pushouts. Applying it to $0 \amalg_{\mathbb{S}} 0 = \Sigma\mathbb{S}$ gives an equivalence of augmented commutative algebras

$$\mathrm{Bar}(\mathrm{Sym}_{\mathbb{S}}(\mathbb{S})) \simeq \mathrm{Sym}_{\mathbb{S}}(\Sigma\mathbb{S}).$$

Let y be the weight-one bottom class in its mod-two homology. Its degree is one, its degree-zero homology is $\mathbb{F}_2$, and there is no negative homology. The counit identities therefore force $\Delta y = 1 \otimes y + y \otimes 1$ for the canonical bar coproduct.

Every power y^n is nonzero. Indeed the weight- n summand is

$$D_n(\Sigma\mathbb{S}) = ((\Sigma\mathbb{S})^{\otimes n})_{h\Sigma_n},$$

whose bottom mod-two homology is $H_n(D_n(\Sigma\mathbb{S}); \mathbb{F}_2) = \mathbb{F}_2$. The n -fold product of weight-one bottom classes maps to this generator. It is induced by the map to homotopy orbits, whose degree-zero base homology map sends one to one; no factorial transfer is involved.

Successor and limit stages. Suppose that the specified augmentation is the composite of a continuous ordinal chain

$$A_0 = \mathrm{Sym}_{\mathbb{S}}(\mathbb{S}) \longrightarrow A_1 \longrightarrow \cdots \longrightarrow A_\lambda = \mathbb{S}$$

whose successor maps are Adams-type and commutative. The maps to the terminal term equip every stage with a compatible sphere augmentation. By Theorem 8.1, the image of y remains primitive and nonnilpotent at a successor stage.

At a nonzero limit ordinal β , the augmented bar functor and mod-two homology preserve the filtered colimit. If any particular power of the image of y were zero at β , that finite relation would already hold at an earlier stage. Hence the image remains nonnilpotent at every limit as well. This argument needs no countable cofinality hypothesis.

The terminal stage. Transfinite induction now gives a nonnilpotent degree-one class at the terminal stage. But $\mathrm{Bar}(\mathbb{S}) \simeq \mathbb{S}$ has zero degree-one mod-two homology. This contradiction proves the theorem and Theorem B(ii). $\square$

The commutative structure of every intermediate map is used in the relative Hopf filtration and in the power operations. Theorem 9.1 makes no assertion about factorizations of this same augmentation through noncommutative intermediate rings. The associative statement of Theorem B is supplied by its separate triangular example.

10 Two-step associative factorizations

Failure of the Adams-type condition does not by itself obstruct a finite Adams-type factorization. We give a triangular construction and determine the minimum lengths for the earlier examples. The intermediate ring is associative and need not be commutative.

Theorem 10.1. *Let $f : A \rightarrow B$ be a map of E_1 -ring spectra. It is a composite of two Adams-type maps in either of the following situations:*

- (i) *f admits an E_1 -ring section $s : B \rightarrow A$, with $fs \simeq \text{id}_B$;*
- (ii) *f is a map of associative Hk -algebras for a field k .*

No connectivity or boundedness assumption is imposed in either case.

The section in (i) is a right inverse to f . This differs from the retraction possessed by the diagonal inclusion in Theorem 5.1. For (ii), the hypothesis means that A and B are algebra objects in Mod_{Hk} and that f respects these structures. An arbitrary map of ring spectra supplied with an unrelated map $Hk \rightarrow B$ is not part of this statement.

Lemma 10.2. *Let A, B be E_1 -rings and J an (A, B) -bimodule. The projection*

$$q : T = \begin{pmatrix} A & J \\ 0 & B \end{pmatrix} \longrightarrow B$$

is Adams-type. Moreover, $B \otimes_T B \simeq B$.

Proof. The triangular ring is the canonical square-zero extension of $A \times B$ by the bimodule supported in its $(1, 2)$ corner. Let e_1, e_2 be its diagonal idempotents and put $P_1 = Te_1$, $P_2 = Te_2$. These are perfect left T -modules. There is an actual cofiber sequence of left T -modules

$$P_1 \otimes_A J \longrightarrow P_2 \longrightarrow B. \quad (10.1)$$

Indeed, the first term is the column $(J, 0)$ and the second is (J, B) , with the triangular actions. In the first term the right A -action on P_1 is the corner action $e_1Te_1 = A$.

Choose a filtered presentation $J \simeq \text{colim}_\alpha J_\alpha$ by perfect left A -modules. Such a presentation exists because the module category is compactly generated by A , with compact objects the perfect modules; see [HA, Proposition 7.2.4.2]. We require no right B -action on the individual J_α . Using their maps to J , form

$$K_\alpha = \text{cofib}(P_1 \otimes_A J_\alpha \longrightarrow P_2). \quad (10.2)$$

The functor $P_1 \otimes_A -$ takes A to the perfect module P_1 and preserves finite cofibers and retracts. Thus every K_α is perfect over T , and (10.1) gives $\text{colim}_\alpha K_\alpha \simeq B$.

For the required base change, it is important to use the right T -module $B \simeq e_2T$. It gives

$$B \otimes_T P_1 \simeq e_2Te_1 = 0, \quad B \otimes_T P_2 \simeq e_2Te_2 = B.$$

Consequently $B \otimes_T K_\alpha \simeq B$ for every α . The modules in (10.2) are therefore Adams-type witnesses. The same right-module identification gives $B \otimes_T B \simeq e_2B \simeq B$. $\square$

The next construction provides the first arrow. It uses a section of a bimodule map rather than a section of the original ring map.

Proposition 10.3. *Let $f : A \rightarrow B$ be a map of E_1 -rings. Suppose that E is an (A, B) -bimodule and that*

$$\mu : E \longrightarrow B$$

is a bimodule map, where the left A -action on B is induced by f . Suppose that μ has a right B -module section $s_0 : B \rightarrow E$, and put $J = \text{fib}(\mu)$. There is a factorization by actual E_1 -ring maps

$$A \xrightarrow{j} T = \begin{pmatrix} A & J \\ 0 & B \end{pmatrix} \xrightarrow{q} B \quad (10.3)$$

of f , for which

$${}_AT \simeq A \oplus E. \quad (10.4)$$

Here the left action is restriction along j . The map q is Adams-type.

Proof. The fiber arrow $J \rightarrow E$ is an object of the stable category of arrows in right B -modules. Its endomorphism spectrum is an E_1 -ring

$$K = \text{End}_{\text{Fun}(\Delta^1, \text{RMod}_B)}(J \rightarrow E).$$

The actual left A -actions give a ring map $A \rightarrow K$, compatible with the map $A \rightarrow \text{End}_B(J)$ obtained by evaluating at the source. Define

$$T = A \times_{\text{End}_B(J)} K \quad (10.5)$$

as a pullback of E_1 -rings. The two compatible maps from A give $j : A \rightarrow T$. Applying the cofiber functor to the arrow $J \rightarrow E$ gives a ring map $K \rightarrow \text{End}_B(B) \simeq B$ and hence $q : T \rightarrow B$. Its composite with j is the original action f .

The chosen right B -module section gives an equivalence of arrows

$$(J \rightarrow E) \simeq (J \xrightarrow{\text{id}} J) \oplus (0 \rightarrow B).$$

Write $X = (J \xrightarrow{\text{id}} J)$ and $Y = (0 \rightarrow B)$ for these two summands. Their mapping spectra in the arrow category are

$$\begin{aligned} \text{Map}(X, X) &\simeq \text{End}_B(J), & \text{Map}(Y, X) &\simeq \text{Map}_B(B, J), \\ \text{Map}(X, Y) &\simeq 0, & \text{Map}(Y, Y) &\simeq \text{End}_B(B). \end{aligned}$$

The zero entry is the fiber of the identity of $\text{Map}_B(J, B)$; this calculation includes the commutativity homotopy in a map of arrows. Composition therefore identifies the endomorphism E_1 -ring with

$$K \simeq \begin{pmatrix} \text{End}_B(J) & \text{Map}_B(B, J) \\ 0 & \text{End}_B(B) \end{pmatrix}.$$

Since $\text{Map}_B(B, J) \simeq J$ and $\text{End}_B(B) \simeq B$, the pullback (10.5) is the triangular ring in (10.3).

Its first column, restricted along j , is the regular left A -module. Its second column is $J \oplus B$ with the action transported from the original left A -action on E . The equivalence $J \oplus B \rightarrow E$ supplied by s_0 identifies that column with E as a left A -module. The possible failure of s_0 to be left A -linear is recorded in the upper-right entry of j . This proves (10.4). Lemma 10.2 proves the assertion about q . $\square$

Proof of Theorem 10.1. Suppose first that f has a ring section s . Use $E = A$, with its regular left A -action and right B -action through s , and take $\mu = f$ and $s_0 = s$. Proposition 10.3 then gives ${}_AT \simeq A \oplus A$. Hence T is perfect as a left A -module and

$$T \otimes_A T \simeq T \oplus T$$

as a left T -module. The single object T is an Adams-type witness for j . The projection q is Adams-type by the proposition.

Now let f be a map of associative Hk -algebras, with k a field. Set

$$E_0 = A \otimes_{Hk} B, \quad \mu(a \otimes b) = f(a)b, \quad s_0(b) = 1 \otimes b. \quad (10.6)$$

Here μ is an actual (A, B) -bimodule map induced by f , and the unit of A induces the right B -module section s_0 . We do not require s_0 to be left A -linear or μ to be a ring map. Proposition 10.3 produces $j : A \rightarrow T$ with ${}_AT \simeq A \oplus E_0$.

Choose a homogeneous basis of the graded vector space π_*B . Representing its elements gives an equivalence of Hk -modules

$$B \simeq \bigoplus_{\nu} \Sigma^{d_{\nu}} Hk.$$

It follows that $E_0 \simeq \bigoplus_{\nu} \Sigma^{d_{\nu}} A$ as a left A -module. Together with (10.4), this expresses T as a coproduct of shifts of A . Finite subcoproducts give perfect left A -module approximations to T whose base changes to T are finite coproducts of shifts of T . Thus j is Adams-type. Again q is Adams-type by Lemma 10.2. $\square$

For the differential graded models used in the examples, the first arrow also has an explicit matrix formula. We record it to identify the factorization without suppressing the twist in its left module structure. In the split case put $J = \ker(f)$ and

$$\delta(a) = a - sf(a), \quad j(a) = \begin{pmatrix} a & \delta(a) \\ 0 & f(a) \end{pmatrix}.$$

The right B -action on J is through s , and

$$\delta(aa') = a\delta(a') + \delta(a)s(f(a')).$$

The map δ is a degree-zero chain map with $\delta(1) = 0$, so this formula is a unital dg algebra map. The second column has action

$$a \cdot (z, b) = (az + \delta(a)s(b), f(a)b),$$

and $(z, b) \mapsto z + s(b)$ identifies it with the regular left A -module.

For a dg k -algebra map without a ring section, use $J_0 = \ker(E_0 \xrightarrow{\mu} B)$ in (10.6) and instead put

$$\delta_f(a) = a \otimes 1 - 1 \otimes f(a), \quad j_f(a) = \begin{pmatrix} a & \delta_f(a) \\ 0 & f(a) \end{pmatrix}. \quad (10.7)$$

Direct cancellation gives $\delta_f(aa') = a\delta_f(a') + \delta_f(a)f(a')$. The second column is identified with E_0 by $(z, b) \mapsto z + 1 \otimes b$, which intertwines the twisted A -action with the original action on E_0 . These formulas compute the ring maps in Proposition 10.3; their spectral coherences were provided by the endomorphism construction, not inferred from an identity on homotopy groups. The dg models and their algebra spectra are related by [Shi, Theorem 1.1].

Corollary 10.4. *Each map in Theorem 3.1 and Propositions 3.5 and 3.7 has minimum finite Adams-type factorization length equal to two in the category of E_1 -ring spectra.*

Proof. The original and Laurent dg augmentations have their displayed ring sections, so the split case of Theorem 10.1 applies to them. The periodic example with connective source is a map of associative $H\mathbb{F}_2$ -algebras, so the field case of Theorem 10.1 applies; its first arrow is given by (10.7) in the dg model. This supplies length-two factorizations in all three cases. The cited results show that each original map is not Adams-type, which excludes length one. An equivalence is Adams-type, so length zero is excluded as well. $\square$

These factorizations separate failure of the Adams-type condition from failure of transfinite Adams-type factorization. They do not extend Theorem 10.1 to arbitrary maps over the sphere, and the triangular intermediate ring does not give a factorization through commutative ring spectra.

11 Positive results for discrete targets and strict differential graded algebras

We give perfect-module approximations for almost perfect modules over a connective source with a discrete coherent target, and countable Adams-type factorizations for maps supplied with strict commutative differential graded models.

11.1 Almost perfect modules and discrete coherent targets

An ordinary commutative ring S is coherent if its finitely generated ideals are finitely presented. In particular, Noetherian rings are coherent. Over a coherent ring, kernels and cokernels of maps between finitely presented modules are finitely presented [Stacks, Tag 05CU].

Theorem 11.1. *Let $A \rightarrow HS$ be a map of associative ring spectra, where A is connective and S is an ordinary commutative coherent ring. Let M be an almost perfect left A -module. Suppose that*

$$\pi_*(HS \otimes_A M)$$

is flat over S , regarded as a graded ring concentrated in degree zero. Then there is a sequence of perfect A -modules $M^{(n)}$ with

$$M \simeq \operatorname{colim}_{n < \omega} M^{(n)},$$

such that $\pi_(HS \otimes_A M^{(n)})$ is a finitely generated projective graded S -module for every n . No Adams-type hypothesis on $A \rightarrow HS$ is required.*

We construct the approximating objects as finite simplicial skeleta over A . After base change, discreteness of the target identifies them with brutal truncations of a complex of finite projective S -modules. The top homology of each truncation consists of cycles in that complex, so the proof must show that these cycles are projective as well as its homology groups.

Proof. After shifting, assume that M is connective. By [HA, Proposition 7.2.4.11(5)], there is a simplicial A -module $P_\bullet$ with

$$P_j \simeq A^{r_j}, \quad r_j < \infty, \quad |P_\bullet| \simeq M.$$

Set

$$M^{(n)} = |\operatorname{sk}_n P_\bullet|. \tag{11.1}$$

The skeletal filtration constructs $M^{(n)}$ from the finite free modules in degrees at most n by finite colimits. Thus each $M^{(n)}$ is perfect, and their colimit is M .

Extension of scalars preserves this filtration. Each $HS \otimes_A P_j$ is a finite free HS -module concentrated in degree zero. The mapping spaces between such modules are discrete, so $HS \otimes_A P_\bullet$ is represented by an ordinary simplicial S -module. Let $D_\bullet$ be its normalized chain complex. Every D_i is a direct summand of a finite free S -module, and

$$H_i(D_\bullet) = \pi_i(HS \otimes_A M).$$

Under the Dold–Kan equivalence, the base change of (11.1) is represented by the *brutal* truncation

$$\sigma_{\leq n} D_\bullet : 0 \longrightarrow D_n \longrightarrow D_{n-1} \longrightarrow \cdots \longrightarrow D_0 \longrightarrow 0.$$

This truncation retains every term through D_n and has no term above degree n .

Put $Z_i = \ker(d : D_i \rightarrow D_{i-1})$ and $B_i = \operatorname{im}(d : D_{i+1} \rightarrow D_i)$. The top homology of the truncation is therefore Z_n . We prove that all cycles and boundaries are finitely generated projective by splitting the two defining exact sequences in each degree. First, coherence implies that $H_i(D_\bullet)$ is finitely presented. It is flat by hypothesis, hence finitely generated projective [Stacks, Tag 00NX, Lemma 10.78.2]. For $i = 0$, one has $Z_0 = D_0$, and

$$0 \longrightarrow B_0 \longrightarrow Z_0 \longrightarrow H_0(D_\bullet) \longrightarrow 0$$

splits. Inductively, projectivity of B_{i-1} splits

$$0 \longrightarrow Z_i \longrightarrow D_i \longrightarrow B_{i-1} \longrightarrow 0,$$

so Z_i is finitely generated projective. Projectivity of $H_i(D_\bullet)$ then splits the sequence with kernel B_i . This proves the claim.

Consequently

$$H_i(\sigma_{\leq n} D_\bullet) = \begin{cases} H_i(D_\bullet), & 0 \leq i < n, \\ Z_n, & i = n, \\ 0, & \text{otherwise.} \end{cases}$$

This is a finitely generated projective graded S -module. The modules in (11.1) therefore have the required properties. Shifting them back proves the assertion for the original M . $\square$

Corollary 11.2. *Let $A \rightarrow HS$ satisfy the ring hypotheses of Theorem 11.1, and suppose that HS is almost perfect as a left A -module. Then $A \rightarrow HS$ is Adams-type if and only if it is descent-flat.*

Proof. For an Adams-type map, a defining presentation of HS becomes, after base change, a filtered presentation of $HS \otimes_A HS$ by homotopy-projective HS -modules. Its homotopy is therefore flat over S . Conversely, apply Theorem 11.1 to $M = HS$. $\square$

Remark 11.3. For example, the almost perfectness condition in Corollary 11.2 holds when A is connective, $\pi_0 A$ is left Noetherian, every $\pi_i A$ is finitely generated over $\pi_0 A$, and S is finitely generated as a left $\pi_0 A$ -module. Indeed, A is then left coherent, and [HA, Proposition 7.2.4.17] applies to the module HS . If S is also a commutative Noetherian ring, all hypotheses of the corollary hold. Neither this criterion nor Theorem 11.1 covers the periodic target in Theorem 4.5.

11.2 Strict differential graded models

Fix an ordinary commutative coefficient ring k . In this subsection, a *strictly graded-commutative DG k -algebra* means a homologically graded algebra with a differential of degree -1 , the Koszul commutativity relation

$$ab = (-1)^{|a||b|}ba,$$

and the additional requirement that every odd-degree homogeneous element has square zero. This last requirement is imposed even when k has 2-torsion. Degrees may be arbitrary integers. We apply the compatible realization $\mathcal{H}$ of Proposition 2.3 to the entire diagram of strict algebras and maps constructed below.

Theorem 11.4. *Let $f : C \rightarrow D$ be a map of strictly graded-commutative DG k -algebras in the preceding sense. There is a diagram*

$$C \longrightarrow C_0 \longrightarrow C_1 \longrightarrow \cdots \longrightarrow D$$

whose maps to D are compatible and whose realization satisfies

$$\operatorname{colim}_{n < \omega} \mathcal{H}(C_n) \simeq \mathcal{H}(D) \quad \text{under } \mathcal{H}(C).$$

Every map $\mathcal{H}(C) \rightarrow \mathcal{H}(C_0)$ and $\mathcal{H}(C_n) \rightarrow \mathcal{H}(C_{n+1})$ is an Adams-type map of commutative ring spectra. There is no boundedness assumption.

Each stage may adjoin an arbitrary set of variables. To prove that such a stage is Adams-type, we exhaust it by finite monomial boxes and show that each box becomes free after base change. We then show that realization preserves filtered colimits, so that the countable construction in DG algebras gives the required spectral factorization.

Lemma 11.5. *Let C be as above. Adjoin a set of homogeneous variables x_i , polynomial in even degrees and exterior in odd degrees, with*

$$|x_i| = n_i, \quad dx_i = a_i \in C_{n_i-1}, \quad da_i = 0.$$

Write D for the resulting strictly graded-commutative DG algebra. Then $\mathcal{H}(C) \rightarrow \mathcal{H}(D)$ is Adams-type.

Proof. The prescribed differential respects the defining relations. In particular, if x_i is odd, then a_i is even and

$$d(x_i^2) = a_i x_i - x_i a_i = 0.$$

Differentiating a commutation relation gives zero by the same Koszul sign rule. Since $da_i = 0$, the square of the differential vanishes on the generators and therefore on the whole algebra. The resulting algebra retains the odd-square-zero convention: the square of an odd monomial vanishes, and the cross terms in the square of a sum of odd monomials cancel.

Choose a finite set J of variables. Give each even variable in J a finite exponent bound N_i ; odd exponents are restricted to 0 and 1. Let $F_{J,N}$ be the C -submodule spanned by the corresponding monomials. It is a subcomplex, because every new variable has differential in C : differentiating a monomial lowers one variable's exponent and leaves the other exponents unchanged. All terms therefore remain within the same bounds. Filtering by the total number of new variables exhibits $F_{J,N}$ as a finite semifree C -module. It is therefore perfect and K -flat. This applies on either side, using graded commutativity, without any flatness assumption on the underlying k -modules of C .

The pairs (J, N) , ordered by inclusion and increasing bounds, form a filtered system, and

$$D = \operatorname{colim}_{(J,N)} F_{J,N} \tag{11.2}$$

as a DG C -module. The boxes are submodules spanned by bounded sets of monomials. Quotienting by large powers would not in general give a differential-stable truncation.

Since $F_{J,N}$ is semifree, its ordinary tensor product computes derived base change. In $D \otimes_C F_{J,N}$, distinguish the variable x_i in the coefficient algebra D from the variable x'_i in the second factor. Set

$$\theta_i = x'_i - x_i.$$

The equality $dx'_i = dx_i = a_i$ gives $d\theta_i = 0$. For odd x_i , Koszul commutativity also gives $\theta_i^2 = 0$. The monomials in the θ_i with the same exponent bounds give a new D -basis. For one even variable, the change of basis is

$$(x' - x)^j = \sum_{\ell=0}^j \binom{j}{\ell} (x')^\ell (-x)^{j-\ell}.$$

Every second-factor exponent on the right is at most j , so the transformation stays within the chosen box. It is triangular with diagonal entries equal to 1. For several variables, ordering monomials by total exponent gives the same conclusion; reordering odd variables introduces only the prescribed signs. The transformation is therefore invertible over any k , without division by integers.

Every element of this new basis is a cycle, so the differential acts only on its D -coefficients. Consequently there is an equivalence of DG D -modules

$$D \otimes_C^{\mathbb{L}} F_{J,N} \simeq \bigoplus_{\alpha} \Sigma \sum_{i \in J} \alpha_i n_i D, \quad (11.3)$$

where the finite sum runs over the chosen exponent box. Proposition 2.3 transports (11.2) and (11.3) to a presentation by perfect $\mathcal{H}(C)$ -modules with finite free base change. This is the Adams-type condition. $\square$

Lemma 11.6. *For a filtered diagram C_{α} of strict DG algebras as above, there is a natural equivalence*

$$\operatorname{colim}_{\alpha} \mathcal{H}(C_{\alpha}) \simeq \mathcal{H}\left(\operatorname{colim}_{\alpha} C_{\alpha}\right)$$

of commutative Hk -algebras. The same filtered colimit is computed in commutative ring spectra.

Proof. Use the operad $\mathcal{O} \rightarrow \operatorname{Com}$ fixed in Proposition 2.3, and regard the entire diagram as a diagram of $\mathcal{O}$ -algebras. This places all objects and maps in the same comparison with commutative Hk -algebras. Filtered colimits of $\mathcal{O}$ -algebras are created on the underlying chain complexes: each operadic operation has finite arity, so its inputs occur together at some stage. Exactness of filtered colimits of modules then implies that they preserve quasi-isomorphisms, including for unbounded complexes.

Take a projectively cofibrant replacement of the diagram in $\mathcal{O}$ -algebras. Its ordinary colimit computes the derived colimit. The map from this ordinary colimit to that of the original diagram is a quasi-isomorphism, by the preceding exactness statement. Thus the strict filtered colimit already represents the colimit in the associated infinity-category. The comparison [RS, Corollary 8.3] preserves that colimit. Finally, filtered colimits of commutative algebra spectra are computed on their underlying spectra [HA, Proposition 3.2.3.1 and Corollary 3.2.3.2]; forgetting the Hk -algebra structure preserves them. This argument uses no model structure on strict commutative DG algebras over k . $\square$

Proof of Theorem 11.4. We use two kinds of variable additions. One initial batch makes the map surjective on homology. Each subsequent batch kills all kernel classes present at that stage; any new kernel classes are handled at the next stage.

For every homogeneous cycle $z \in D$, adjoin to C a closed variable u_z of degree $|z|$, and send u_z to z . Use a polynomial variable in even degree and an exterior variable in odd degree. The latter assignment is legitimate because every odd z has square zero in D . This gives

$$C \longrightarrow C_0 \xrightarrow{f_0} D$$

with $H_*(f_0)$ surjective. Lemma 11.5 proves that the first map is Adams-type after realization.

Suppose that $f_n : C_n \rightarrow D$ has been constructed and is surjective on homology. For each homogeneous element of $\ker H_*(f_n)$, choose a cycle representative $a \in C_n$ and an element $b \in D$ with $db = f_n(a)$. Adjoin a variable v_a with

$$|v_a| = |a| + 1, \quad dv_a = a, \quad f_{n+1}(v_a) = b.$$

If v_a is odd, so is b , and $b^2 = 0$. Hence this extends f_n to a strict DG map f_{n+1} . All the new differentials lie in the old algebra C_n , so Lemma 11.5 applies to the entire batch. Surjectivity on homology persists: an old cycle mapping to a nonzero class in D cannot become a boundary in C_{n+1} .

Let $C_\infty = \operatorname{colim}_{n < \omega} C_n$. Exactness of filtered colimits gives

$$H_*(C_\infty) \cong \operatorname{colim}_n H_*(C_n).$$

This homology maps surjectively to $H_*(D)$. A class in its kernel is represented at a finite stage C_n , where it already maps to zero in $H_*(D)$; it is killed in C_{n+1} . It follows that $C_\infty \rightarrow D$ is a quasi-isomorphism. Thus countably many stages suffice, independently of the number of variables in each batch. Lemma 11.6 now identifies the realized colimit with $\mathcal{H}(D)$. Every map to D in the construction is compatible with the original f , so the resulting equivalence is under $\mathcal{H}(C)$. $\square$

Example 11.7. Over $\mathbb{Z}$, take $C = \Lambda_{\mathbb{Z}}(e)$, with $|e| = 1$ and $de = 0$, and adjoin an even variable t with $|t| = 2$ and $dt = e$. Then

$$\begin{aligned} d(t^m) &= met^{m-1}, & H_0(C[t]) &= \mathbb{Z}, \\ H_{2m-1}(C[t]) &= \mathbb{Z}/m, & H_{2m}(C[t]) &= 0 \quad (m \geq 1). \end{aligned}$$

Thus adjoining a variable can create new homology that must be handled at later stages. It does not invalidate the Adams presentation: the finite submodules spanned by $1, t, \dots, t^N$ have free base change with basis $1, t' - t, \dots, (t' - t)^N$.

Remark 11.8. Theorem 11.4 concerns maps with the specified strict DG models. It does not strictify arbitrary commutative ring spectra. In particular, the comparison used here is with E_∞ algebras in unbounded chain complexes [RS, Corollary 8.3]; the strict commutative comparison in [RS, Corollary 8.4] has a rational coefficient hypothesis. The odd-square-zero convention also matters for the construction: over $\mathbb{F}_2$, a polynomial generator of odd degree with nonzero square cannot be the image of one of our exterior variables. Finally, the argument proves a countable composite of Adams-type maps, without claiming that this composite is itself Adams-type.

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The results used here include Tags 023N, 05CU, and 00NX.